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A FAST TEMPLATE PERIODOGRAM FOR DETECTING NON-SINUSOIDAL FIXED-SHAPE SIGNALS IN IRREGULARLY SAMPLED TIME SERIES

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ABSTRACT

Astrophysical time series often contain periodic signals. The large and growing volume of time series data from photometric surveys demands computationally efficient methods for detecting and characterizing such signals. The most efficient algorithms available for this purpose are those that exploit the $\mathcal{O}(N \log N)$ scaling of the Fast Fourier Transform (FFT) in the transform size N. However, these methods are not optimal for non-sinusoidal signal shapes. Template fits (or periodic matched filters) optimize sensitivity for *a priori* known signal shapes—for example, the characteristic lightcurves of RR Lyrae, Cepheids, δ Scuti variables, eclipsing binaries, and exoplanetary transits—but at a significant computational cost. Current implementations of template periodograms scale as $\mathcal{O}(N_f N_{\rm obs})$, where N_f is the number of trial frequencies and $N_{\rm obs}$ is the number of lightcurve observations, and due to non-convexity, they do not guarantee the best fit at each trial frequency, which can lead to spurious results. In this work, we present a non-linear extension of the Lomb-Scargle periodogram—non-linear in the sense that the joint optimization over the template's phase shift and harmonic amplitudes is non-convex, even though the model is linear in the amplitudes at any fixed phase shift—to obtain a template-fitting algorithm that is both accurate (globally optimal solutions are obtained except in pathological cases—in practice, when a single observation dominates the total inverse-variance weight budget by a factor $\gtrsim 10^8$) and computationally efficient (scaling as $\mathcal{O}(N_f \log N_f)$ for a given template). The non-linear optimization of the template fit at each frequency is recast as a polynomial zero-finding problem, where the coefficients of the polynomial can be computed efficiently with the non-equispaced fast Fourier transform. We show that our method, which uses truncated Fourier series to approximate templates, is more than an order of magnitude faster than existing non-linear optimization algorithms even for the smallest problems tested ($\approx 14\times$ at $N_{\rm obs} = 15$ observations) and more than two orders of magnitude faster ($\gtrsim 480\times$) by $N_{\rm obs} = 250$, an advantage that grows without bound as $N_{\rm obs}$ increases. Signals with very sharp features (e.g. limb-darkened transits with brief ingress, sharp eclipses) require moderate-to-high harmonic orders ($H \gtrsim 20$) to be reproduced at the 99% L^2 level (Table 1); for such signals, transit-tailored methods such as the Box Least-Squares periodogram may be more efficient. An open-source implementation of the fast template periodogram is freely available online.[a]

KEYWORDS

methods: data analysis – methods: statistical – techniques: photometric – stars: variables: general – stars: variables: RR Lyrae

1. INTRODUCTION

Astronomical systems exhibit a wide range of time-dependent variability. By measuring and characterizing this variability, astronomers are able to infer a variety of important astrophysical properties about the underlying system.

Periodic signals in noisy astronomical timeseries can be detected with a number of techniques, including Gaussian process regression (Foreman-Mackey et al. 2017; Rasmussen & Williams 2005; Aigrain & Foreman-Mackey 2023; Gordon et al. 2021), least-squares spectral analysis (Lomb 1976; Scargle 1982; Barning 1963; Vaníček 1971), and information-theoretic methods (Graham et al. 2013a; Huijse et al. 2012; Cincotta et al. 1995). For an empirical comparison of some of these techniques applied to several astronomical survey datasets, see Graham et al. (2013b).

For stationary periodic signals, least-squares spectral analysis — also known as the Lomb-Scargle (LS) periodogram (Lomb 1976; Scargle 1982; Barning 1963; Vaníček 1971) — is perhaps the most sensitive and computationally efficient method of detection. The LS periodogram can be made to scale as $\mathcal{O}(N_f \log N_f)$, where N_f is the number of trial frequencies, by utilizing the non-equispaced fast Fourier transform (Keiner et al. 2009; Dutt & Rokhlin 1993, NFFT) to evaluate frequency-dependent sums, or by "extirpolating" irregularly spaced observations to a regular grid with Lagrange polynomials (Press & Rybicki 1989).

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[a] https://github.com/PrincetonUniversity/FastTemplatePeriodogram

The LS periodogram fits the following model to a set of observations:

$$\hat{y}_{\rm LS}(t;\theta,\omega) = \theta_0 \cos\omega t + \theta_1 \sin\omega t. \tag{1}$$

where $\omega = 2\pi/P$ is the (angular) frequency of the underlying signal, and θ_0 and θ_1 are the amplitudes of the signal. When the data y_i is composed of a sinusoidal component and a white noise component (i.e., when the measurement uncertainties are uncorrelated and Gaussian), the LS periodogram provides a maximum likelihood estimate for the model parameters (ω, θ_0, and θ_1).

The LS "power" $P_{LS}(\omega)$ has several definitions in the literature (Zechmeister & Kürster 2009), but we adopt the following definition throughout the paper:

$$P(\omega) = \frac{V_0 - V(\omega)}{V_0} \tag{2}$$

where V_0 is the weighted sum of squared residuals for a constant fit:

$$V_0 = \sum_i \mathsf{w}_i (y_i - \bar{y})^2 \tag{3}$$

with $\bar{y} = \sum_i \mathsf{w}_i y_i$ the weighted mean of the observations and $\mathsf{w}_i \propto \sigma_i^{-2}$ the normalised weights for each observation ($\sum_i \mathsf{w}_i = 1$), and $V(\omega)$ is the weighted sum of squared residuals for the best-fit model $\hat{y}$ at a given trial frequency $\omega = 2\pi/P$ where P is the period:

$$V(\omega) = \min_\theta \sum_i \mathsf{w}_i (y_i - \hat{y}(t_i;\omega,\theta))^2. \tag{4}$$

We use the symbol V rather than χ^2 to follow the convention common in the linear least-squares literature: with $\sum_i \mathsf{w}_i = 1$ the quantity $\sum_i \mathsf{w}_i (y_i - \hat{y}_i)^2$ is the weighted sum of squared residuals but is not strictly χ^2-distributed, so the χ^2 symbol can be misleading.

1.1. *Maximum-likelihood interpretation*

Although the periodogram in Equation 2 is most commonly introduced as a least-squares fit, it has a direct probabilistic reading. Under the standard noise assumptions made throughout this paper – uncorrelated, zero-mean, Gaussian uncertainties $y_i \sim \mathcal{N}(\hat{y}(t_i;\omega,\theta), \sigma_i^2)$ – the likelihood of the data given the model parameters is

$$p(\mathcal{D}|\omega,\theta) = \prod_{i=1}^{N_{\rm obs}} \mathcal{N}(\hat{y}(t_i;\omega,\theta), \sigma_i^2), \tag{5}$$

where we use $\mathcal{D} = \{(t_i, y_i, \sigma_i)\,|\, 0 < i \le N_{\rm obs}\}$ as shorthand for the lightcurve observations (avoiding the symbol X, which is conventionally reserved for the design matrix). The log-likelihood is

$$\log p(\mathcal{D}|\omega,\theta) = -\frac{1}{2}\sum_{i=1}^{N_{\rm obs}} \left(\frac{y_i - \hat{y}(t_i;\omega,\theta)}{\sigma_i}\right)^2 + \text{const.} \tag{6}$$

$$= -\frac{W}{2} V(\theta,\omega) + \text{const.}, \tag{7}$$

where $W = \sum_i \sigma_i^{-2}$ and $V(\theta,\omega) \equiv \sum_i \mathsf{w}_i (y_i - \hat{y}(t_i;\omega,\theta))^2$ is the weighted sum of squared residuals written with the normalised weights $\mathsf{w}_i = \sigma_i^{-2}/W$ ($\sum_i \mathsf{w}_i = 1$). Hence

$$V(\omega) = \min_\theta \sum_i \mathsf{w}_i (y_i - \hat{y}(t_i;\omega,\theta))^2 \tag{8}$$

$$= -\frac{2}{W} \max_\theta \log p(\mathcal{D}|\omega,\theta) + \text{const.}, \tag{9}$$

and therefore, since $P(\omega) = 1 - V(\omega)/V_0$,

$$P(\omega) = \frac{2}{WV_0} \max_\theta \log p(\mathcal{D}|\omega,\theta) + \text{const.} \tag{10}$$

The Lomb-Scargle power is therefore a linear, monotonically increasing transformation of the profile log-likelihood – that is, of the log-likelihood maximised over the non-frequency parameters θ at each trial ω. Choosing the frequency that maximises the periodogram is equivalent to selecting the maximum-likelihood frequency estimate under the assumed noise model.

It is tempting to recast this as a maximum-a-posteriori (MAP) statement under an implicit uniform prior on θ, but we deliberately avoid that framing for two reasons. First, a uniform prior over an unbounded domain is improper; recovering the periodogram as a strict MAP quantity would require a piecewise prior on (θ_0, θ_1) that vanishes outside some explicitly chosen bounding box, which we do not impose. Second, even when bounded, a uniform prior on (θ_0, θ_1) implicitly imposes a linear prior on the sinusoid amplitude $A \equiv \sqrt{\theta_0^2 + \theta_1^2}$, since the prior volume element in the (θ_0, θ_1) plane is $\mathrm{d}\theta_0\,\mathrm{d}\theta_1 = A\,\mathrm{d}A\,\mathrm{d}\phi$. A uniform prior on the Cartesian coefficients therefore biases the inference towards large amplitudes. The same pitfall is recognised in exoplanet eccentricity inference, where the parameterisation $(e\cos\varpi,\, e\sin\varpi)$ – introduced for sampling efficiency by Ford (2006) – implicitly imposes a linear prior on the eccentricity e that has since been corrected for or replaced (Eastman et al. 2013); the prior-choice question remains active in this domain (Stevenson et al. 2025). The maximum-likelihood reading invoked here makes none of these implicit prior assumptions, and we adopt it throughout.

A genuinely Bayesian treatment of the LS periodogram, in which the non-frequency parameters are marginalised rather than profiled, was developed by Mortier et al. (2015a,b). Their periodogram power is

$$P_{\rm Bayes}(\omega) \propto p(\omega|\mathcal{D}) = \int p(\omega,\theta|\mathcal{D})\,\mathrm{d}\theta,$$

evaluated under the same uncorrelated-Gaussian noise model but with explicit, bounded priors on the non-frequency parameters. Mortier et al. (2015a) focus on the classical periodogram model of Lomb (1976) and Scargle (1982) together with the constant-offset extension of Zechmeister & Kürster (2009); the same marginalisation strategy could in principle be applied to the template-based formulation developed in this paper.

As remarked in (VanderPlas 2018), the probabilistic interpretation of a periodogram is only as good as the noise model underlying it; in practice the working assumptions (uncorrelated, Gaussian uncertainties, no systematics) are often broken in real astronomical timeseries, and the maximum-likelihood reading above should be understood with the same caveats. Hara et al. (2022) provide a complementary diagnostic for testing whether a periodogram peak corresponds to a strictly periodic component or to time-varying activity, which is useful when the noise model is uncertain.

1.2. *Extending Lomb-Scargle*

The LS periodogram has numerous extensions to account for, e.g., biased estimates of the mean brightness (Zechmeister & Kürster 2009), non-sinusoidal signals (Schwarzenberg-Czerny 1996; Bretthorst 2003; Palmer 2009; Baluev 2009, 2013c; Wheeler & Kipping 2019), joint matched-filter and Gaussian-process noise models (Garcia et al. 2024), multiple periodicities (Baluev 2013a,b), multi-band observations (VanderPlas & Ivezić 2015), and to mitigate overfitting of more flexible models via regularization (VanderPlas & Ivezić 2015). Baluev (2008) provided the framework for improving false alarm statistics with extreme value theory, and this was generalized further in Baluev (2013a). For a more detailed review of the LS periodogram and its extensions, see VanderPlas (2018).

The multi-harmonic LS periodogram (Bretthorst et al. 1988; Schwarzenberg-Czerny 1996; Palmer 2009, MHGLS), provides a more flexible model by adding harmonic components to the fit:

$$\hat{y}_{\rm MHLS}(t;\theta,\omega) = \theta_0 + \sum_{n=1}^{H} \theta_{2n}\cos n\omega t + \theta_{2n-1}\sin n\omega t. \tag{11}$$

Additional harmonics are important for modeling signals that, while stationary and periodic, are non-sinusoidal (e.g. RR Lyrae, eclipsing binaries, etc.).

However, the MHLS periodogram contains $2H + 1$ free parameters while the original LS periodogram contains only 2 (3 if the mean brightness is considered a free parameter). Including higher order harmonics adds model complexity, which can degrade the sensitivity of the MHLS periodogram to sinusoidal or approximately sinusoidal signals. We emphasise that this is a model-selection / signal-to-noise penalty—more free parameters absorb more noise variance from each trial frequency—rather than an optimisation failure: the MHLS least-squares problem is linear in θ at fixed ω and its global minimum can be obtained directly without iteration.

Tikhonov regularisation (also known as L_2 regularisation) is one tool for mitigating overfitting of the higher-order harmonics. Adding an L_2 penalty on the Fourier amplitudes is equivalent to imposing a zero-mean Gaussian prior on those amplitudes; in the limit of infinite prior width, this prior reduces to the implicit (improper) uniform prior used in the unregularised fit. The choice of prior width is therefore a subjective modelling decision rather than a bias in the statistical sense, and its merits should be weighed against the corresponding decrease in model variance (VanderPlas & Ivezić 2015).

1.3. *Computational scaling*

The LS periodogram naively scales as $\mathcal{O}(N_f N_{\rm obs})$ where N_f is the number of trial frequencies and $N_{\rm obs}$ is the number of observations. However, the limiting computations of the LS periodogram involve sums of trigonometric functions over the observations. When the observations are regularly sampled, the fast Fourier transform (FFT) (Cooley & Tukey 1965) can evaluate such sums efficiently and the LS periodogram scales as $\mathcal{O}(N_f \log N_f)$.

When the data is not regularly sampled, as is the case for most astronomical time series, the LS periodogram can be evaluated quickly in one of two popular ways. The first, by Press & Rybicki (1989) involves "extirpolating" irregularly sampled data onto a regularly sampled mesh, and then performing FFTs to evaluate the necessary sums. The second, as pointed out in Leroy (2012), is to use the non-equispaced FFT (Keiner et al. 2009; Dutt & Rokhlin 1993, NFFT) to evaluate the sums; this provides roughly an order of magnitude speedup over the Press & Rybicki (1989) algorithm, and both algorithms scale as $\mathcal{O}(N_f \log N_f)$ (Leroy 2012). A third option—used in many production pipelines—is to bypass the FFT altogether and solve the per-frequency $2{\times}2$ (or $(2H{+}1){\times}(2H{+}1)$ in the multi-harmonic case) normal equations directly with a dense linear solver; this is simpler to implement than the NFFT-based formulation but loses the $\mathcal{O}(\log N_f)$ scaling. We adopt the NFFT formulation in this work both for its asymptotic speed at the large N_f relevant to upcoming surveys and because, as shown in §2, the NFFT-computable trigonometric sums are precisely the polynomial coefficients required to recast the non-linear template-fitting step as a root-finding problem.

There is a growing population of alternative methods for detecting periodic signals in astrophysical data. Some of these methods can reliably outperform the LS periodogram, especially for non-sinusoidal signal shapes (see Graham et al. (2013b) for a recent empirical review of period finding algorithms). However, a key advantage the LS periodogram and its extensions is speed. Virtually all other "phase-folding" methods scale as $\mathcal{O}(N_{\rm obs} \times N_f)$, where $N_{\rm obs}$ is the number of observations and N_f is the number of trial frequencies, while the Lomb-Scargle periodogram scales as $\mathcal{O}(N_f \log N_f)$. The virtual independence of Lomb-Scargle's computation time with respect to the number of observations (assuming $N_f \gtrsim N_{\rm obs}$) is especially valuable for lightcurves with $N_{\rm obs} \gg \log N_f \sim 50$. To put absolute numbers on this, computing a Lomb–Scargle periodogram over $N_f = 10^4$ trial frequencies takes $\approx 3\,$ms per source with the NFFT formulation, essentially independent of $N_{\rm obs}$ across $N_{\rm obs} = 50$–1000 ($\approx 0.3\,\mu$s per trial frequency); evaluating the same grid with the direct $\mathcal{O}(N_f N_{\rm obs})$ method instead ranges from $\approx 85\,$ms at $N_{\rm obs} = 50$ to $\approx 210\,$ms at $N_{\rm obs} = 1000$ (≈ 9–$21\,\mu$s per trial frequency).[5]

Algorithmic efficiency will become increasingly important as the volume of data produced by astronomical observatories continues to grow larger. The HATNet survey (Bakos et al. 2004), for example, has already made $\mathcal{O}(10^4)$ observations of $\mathcal{O}(10^6 - 10^7)$ stars. The Gaia mission (Gaia Collaboration et al. 2016) has now delivered

[5] Representative single-core timings (`gatspy`, Apple Silicon/`arm64`, Python 3.9, NumPy 2.0) over an explicit $[0.01, 10]\,\mathrm{cyc\,day}^{-1}$ frequency grid; reproduction script at `scripts/timing_v5/ls_timing.py`. Absolute values are hardware-dependent and will be refreshed for the final submission.

$\mathcal{O}(10-100)$ observations of $\mathcal{O}(10^9)$ stars and a dedicated variability processing pipeline that classifies $\mathcal{O}(10^7)$ variables across 35 object types (Eyer et al. 2023). The ASAS-SN survey has likewise contributed $\mathcal{O}(10^5)$ new periodic variables from its all-sky g-band photometry (Christy et al. 2023). The Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST; LSST Science Collaboration et al. 2009), which began science operations in 2025, is expected to make $\mathcal{O}(10^2-10^3)$ observations of $\mathcal{O}(10^{10})$ stars over its ten-year survey, and the cadence-dependent reliability of standard-candle classification on the LSST schedule has been analysed in detail by Hernitschek & Stassun (2021).

1.4. *Template periodograms*

When the shape of a stationary periodic signal is known *a priori*, then the number of degrees of freedom is the same as the original LS periodogram (with a floating mean component):

$$\hat{y}(t;\theta,\omega)=\theta_0+\theta_1\mathbf{M}(\omega t-\theta_2), \tag{12}$$

where $\mathbf{M}:[0,2\pi)\to\mathbb{R}$ is a predefined periodic template (i.e., a function from the unit circle to the real line). We refer to the periodogram corresponding to this model as the "template periodogram."

As is the case for the LS periodogram, the template periodogram is equivalent to a maximum-likelihood estimate of the model parameters under the assumption that measurement uncertainties are Gaussian and uncorrelated (i.e. white noise).

This paper develops new extensions of least-squares spectral analysis for arbitrary signal shapes. For non-periodic signals this method is known as matched filter analysis, and can be extended to search for periodic signals by, e.g., phase folding the data at different trial periods.

An analysis by Sesar et al. (2016) found that template fitting significantly improved period and amplitude estimation for RR Lyrae in Pan-STARRS DR1 photometry (Chambers et al. 2016). Since the signal shapes for RR Lyrae in various bandpasses are known *a priori* (see Sesar et al. (2010)), template fitting provides an optimal estimate of amplitude and period, given that the object is indeed an RR Lyrae star well modeled by at least one of the templates. The Gaia DR3 RR Lyrae and Cepheid samples (Clementini et al. 2023; Ripepi et al. 2023) extend this template-driven approach to all-sky catalogues of $\mathcal{O}(10^5)$ stars, and Forró et al. (2024) report $\sim 90\%$ agreement between Pan-STARRS PS1 template-derived RR Lyrae periods and *K2* high-cadence measurements. Templates were especially crucial for Pan-STARRS data, since there are typically only 35 observations per source over 5 bands (Hernitschek et al. 2016), not enough to obtain accurate amplitudes empirically by phase-folding. By including domain knowledge (i.e. knowledge of what RR Lyrae lightcurves look like), template fitting allows for accurate inferences of amplitude even for undersampled lightcurves.

However, the improved accuracy comes at substantial computational cost: the template fitting procedure took 30 minutes per CPU per object, and Sesar et al. (2016) were forced to limit the number of fitted lightcurves ($\lesssim 1000$) in order to keep the computational costs to a reasonable level. Several cuts were made before the template fitting step to reduce the more than 1 million Pan-STARRS DR1 objects to a small enough number, and each of these steps removes a small portion of RR Lyrae from the sample. Though this number was reported by Sesar et al. (2016) to be small ($\lesssim 2\%$), it may be possible to further improve the completeness of the final sample by applying template fits to a larger number of objects, which would require either more computational resources, more time, or, ideally, a more efficient template fitting procedure.

We note that the fast template periodogram (FTP), like the LS periodogram, assumes that the reported photometric uncertainties σ_i are strictly positive and fully describe the noise. In practice, real photometric data frequently exhibit additional intrinsic scatter (or "equation error") beyond the reported σ_i, so that the model–data residual variance exceeds σ_i^2. In the least-squares context, the FITEXY algorithm (Press et al. 1992, *Numerical Recipes*; see also Press & Rybicki 1989) and the BCES estimators of Akritas & Bershady (1996) accommodate heteroscedastic measurement errors together with a homoscedastic equation-error term, and Kelly (2007) shows that a hierarchical-Bayesian, likelihood-based treatment delivers less bias and higher accuracy than these least-squares approximations (with reference implementations available as `LINMIX_ERR` in IDL, `linmix` in Python, and `lrgs` in R). Extending the FTP to jointly infer an unknown intrinsic scatter would require a likelihood-based generalization beyond the closed-form least-squares structure exploited here and is out of scope for the present paper. As a first-order practical correction, users with evidence of underestimated uncertainties can inflate each σ_i by a jitter term added in quadrature ($\sigma_i^2\to\sigma_i^2+s^2$), as is standard practice in radial-velocity and *TESS* analyses; the FTP machinery is unchanged.

The paper is organized as follows. Section 2 poses the problem of template fitting in the language of least squares spectral analysis and derives the fast template periodogram. Section 3 describes a freely available implementation of the new template periodogram. Section 6 summarizes our results, addresses caveats, and discusses possible avenues for improving the efficiency of the current algorithm.

2. DERIVATIONS

We define a template $\mathbf{M}$

$$\mathbf{M}:[0,2\pi)\to\mathbb{R}, \tag{13}$$

as a mapping between the unit interval and the set of real numbers. We restrict our discussion to sufficiently smooth templates such that $\mathbf{M}$ can be adequately described by a truncated Fourier series

$$\hat{\mathbf{M}}(\omega t;H)=\sum_{n=1}^{H}\left(c_n\cos n\omega t+s_n\sin n\omega t\right) \tag{14}$$

for some finite $H>0$.

That the c_n and s_n values are *fixed* (i.e., they define the template) is the crucial difference between the template periodogram and the multi-harmonic Lomb-Scargle

(Palmer 2009; Bretthorst et al. 1988), where c_n and s_n are *free parameters*.

We now construct a periodogram for this template. The periodogram assumes that an observed time series $S = \{(t_i, y_i, \sigma_i)\}_{i=1}^N$ can be modeled by a scaled, transposed template that repeats with period $2\pi/\omega$, i.e.

$$y_i \approx \hat{y}(\omega t_i; \theta, \mathbf{M}) = \theta_1 \mathbf{M}(\omega t_i - \theta_2) + \theta_3, \qquad (15)$$

where $\theta = (\theta_1, \theta_2, \theta_3) \in \mathbb{R}^3$ is a set of model parameters. The optimal parameters are the location of a local minimum of the (weighted) sum of squared residuals,

$$V(\theta, S) \equiv \sum_i \mathsf{w}_i (y_i - \hat{y}(\omega t_i; \theta))^2, \qquad (16)$$

and thus the following condition must hold for all three model parameters at the optimal solution $\theta = \theta_{\rm opt}$:

$$\left.\frac{\partial V}{\partial \theta_j}\right|_{\theta=\theta_{\rm opt}} = 0 \ \ \forall \theta_j \in \theta. \qquad (17)$$

Note that we have implicitly assumed $V(\theta, S)$ is a C^1 differentiable function of θ, which requires that $\mathbf{M}$ is a C^1 differentiable function. Though this assumption could be violated if we considered a more complete set of templates, (e.g. a box function), our restriction to truncated Fourier series ensures C^1 differentiability.

Note that we also implicitly assume $\sigma_i > 0$ for all i and we will later assume that the variance of the observations y_i is non-zero. If there are no measurement errors, i.e. $\sigma_i = 0$ for all i, then uniform weights (setting $\sigma_i = 1$) should be used. If the variance of the observations y is zero, the periodogram (as defined in Equation 2) is undefined for all frequencies. We do not consider the case where $\sigma_i = 0$ for some observations i and $\sigma_j > 0$ for some observations j.

We can derive a system of equations for $\theta_{\rm opt}$ from Equation 17. The fast template periodogram reduces this three-parameter non-linear minimisation to a polynomial root-finding problem in three steps; the qualitative outline is given here, with the explicit algebra collected in Appendix A.

Step 1: closed form for (θ_1, θ_3) at fixed θ_2. The model $\hat{y}$ is linear in θ_1 and θ_3 once the phase shift θ_2 is held fixed, so the conditions $\partial V/\partial\theta_1 = \partial V/\partial\theta_3 = 0$ form a 2×2 linear system with closed-form solution (Appendix A, Equation A11):

$$\theta_1 = \frac{YM}{MM}, \qquad \theta_3 = \bar{y} - \overline{M}\left(\frac{YM}{MM}\right), \qquad (18)$$

where the weighted statistics

$$\begin{aligned} \overline{M} &\equiv \langle \mathbf{M}_{\theta_2} \rangle, & \bar{y} &\equiv \langle y \rangle, \\ MM &\equiv {\rm Var}(\mathbf{M}_{\theta_2}), & YM &\equiv {\rm Cov}(\mathbf{M}_{\theta_2}, y), \\ YY &\equiv {\rm Var}(y) & & \end{aligned} \qquad (19)$$

are all implicit functions of θ_2 through the phase-shifted template $\mathbf{M}_{\theta_2} \equiv \mathbf{M}(\omega t - \theta_2)$, and the weighted-average operator $\langle\cdot\rangle$, covariance Cov, and variance Var are defined in Appendix A. Substituting Equation 18 back into V and using $V_0 = YY$, the periodogram $P(\omega) \equiv 1 - V/V_0$ takes the compact form

$$P(\omega) = \frac{[YM(\theta_2)]^2}{YY \cdot MM(\theta_2)}. \qquad (20)$$

The intermediate algebra is given in Appendix A.

Step 2: reduce to a one-dimensional optimisation over θ_2. Maximising Equation 20 with respect to the remaining non-linear parameter θ_2 gives

$$\begin{aligned} \partial_{\theta_2} P = 0 &= \frac{YM}{YY \cdot MM}\left(2\partial_{\theta_2}(YM) - \frac{YM}{MM}\partial_{\theta_2}(MM)\right) \\ &= 2\, MM\, \partial_{\theta_2}(YM) - YM\, \partial_{\theta_2}(MM). \end{aligned} \qquad (21)$$

This transcendental condition on the single parameter θ_2 is what existing non-linear template fitters must solve numerically at every trial frequency. Satisfying Equation 21 is necessary but not *sufficient*: the value of $P(\omega)$ must be evaluated at each root, and the global maximum selected from this set.

Step 3: recast as polynomial root-finding. The crux of the FTP is the change of variable $\psi \equiv e^{i\theta_2}$ together with the rescaling $YM' \equiv \psi^H YM$ and $MM' \equiv \psi^{2H} MM$. Because the phase-shifted template (Equation 14) enters YM and MM only through factors of $e^{\pm in\theta_2}$ with $n \le H$, the rescaled quantities YM' and MM' are polynomials in ψ (of degree $2H$ and $4H$, respectively; Appendix A). A short calculation (Appendix A) shows that the non-linear condition Equation 21 is equivalent to

$$0 \;=\; 2\, MM'\, \partial(YM') \;-\; YM'\, \partial(MM'), \qquad (22)$$

a polynomial of degree at most $6H-2$ in ψ: the nominal degree-$(6H-1)$ terms cancel identically, because both products have leading coefficient $4H$ times the product of the leading coefficients of YM' and MM' ($\deg MM' = 2 \deg YM'$). The coefficients of this polynomial are linear combinations of the weighted trigonometric sums

$$CC_{nm} \equiv {\rm Cov}(\cos n\omega t, \cos m\omega t), \qquad (23)$$
$$CS_{nm} \equiv {\rm Cov}(\cos n\omega t, \sin m\omega t), \qquad (24)$$
$$SS_{nm} \equiv {\rm Cov}(\sin n\omega t, \sin m\omega t), \qquad (25)$$
$$YC_n \equiv \langle (y - \bar{y}) \cos n\omega t \rangle, \qquad (26)$$
$$YS_n \equiv \langle (y - \bar{y}) \sin n\omega t \rangle, \qquad (27)$$

each of which can be evaluated at *all* trial frequencies ω simultaneously in $\mathcal{O}(HN_f \log HN_f)$ time using the non-equispaced fast Fourier transform. The explicit assembly of the polynomial coefficients from these sums—and the algebraic identity that converts Equation 21 into Equation 22—is given in Appendix A (Equations A19–A26).

We solve for the roots of Equation 22 using the `numpy.polynomial.polyroots` function, which returns the eigenvalues of the polynomial's companion matrix. Multiplying through by powers of ψ in deriving Equation 22 introduces spurious roots, and ψ must lie on the unit circle ($|\psi| = 1$ for real θ_2). We therefore scale each root by its modulus to enforce the unit-circle constraint; iterative schemes such as Newton's method could be used for greater robustness, but the measured periodogram error floor of the present implementation is $\sim 10^{-15}$ (direct summations) to $\sim 10^{-13}$ (NFFT-based summations), verified through $H = 21$ (Section 2.4). This floor is measured under homoscedastic measurement errors; it degrades when a single observation dominates the total inverse-variance weight budget by a factor $\gtrsim 10^8$, with the fast NFFT-based summations degrading earlier than direct summations (Section 2.4).

Summary of the algorithm. We have derived an explicit, non-linear system of equations to solve for the optimal model parameters at each trial frequency. Concretely, for each ω the FTP proceeds as follows:

1. Compute the weighted trigonometric sums of Equations 23–27 at all trial frequencies ω using the NFFT.
2. Assemble the coefficients of the degree-($\leq 6H-2$) polynomial of Equation 22 from the sums of step 1; the closed-form assembly is given in Appendix A (Equations A19–A23).
3. Find all roots of this polynomial via `numpy.polynomial.polyroots`, retaining only the roots that lie on the unit circle $\psi = e^{i\theta_2}$.
4. For each candidate θ_2, evaluate the corresponding linear-best-fit (θ_1, θ_3) via Equation 18 and the periodogram $P(\omega)$ via Equation 20.
5. Declare the largest $P(\omega)$ value the global optimum; the corresponding $(\theta_1, \theta_2, \theta_3)$ are the best-fit model parameters at that trial frequency.

2.1. *Negative amplitude solutions*

The model $\hat{y} = \theta_1 \mathbf{M}_{\theta_2} + \theta_0$ allows for $\theta_1 < 0$ solutions. In the original formulation of Lomb-Scargle and in linear extensions involving multiple harmonics, negative amplitudes translate to phase differences, since $-\cos x = \cos(x - \pi)$ and $-\sin x = \sin(x - \pi)$.

However, for non-sinusoidal templates, $\mathbf{M}$, negative amplitudes do not generally correspond to a phase difference. For example, a detached eclipsing binary template $\mathbf{M}_{\rm EB}(x)$ cannot be expressed in terms of a phase-shifted negative eclipsing binary template; i.e. $\mathbf{M}_{\rm EB} \neq -\mathbf{M}_{\rm EB}(x - \phi)$ for any $\phi \in [0, 2\pi)$.

Negative amplitude solutions found by the fast template periodogram are usually undesirable, as they may produce false positives for lightcurves that resemble flipped versions of the desired template, and allowing for $\theta_1 < 0$ solutions increases the number of effective free parameters of the model, which lowers the signal to noise, especially for weak signals.

One possible remedy for this problem is to set $P_{\rm FTP}(\omega) = 0$ if the optimal solution for θ_1 is negative, but this complicates the interpretation of $P_{\rm FTP}$. Another possible remedy is, for frequencies that have a $\theta_1 < 0$ solution, to search for the optimal parameters while enforcing that $\theta_1 > 0$, e.g. via non-linear optimization, but this likely will eliminate the computational advantage of FTP over existing methods.

The cost of the positivity constraint is, however, much smaller than the cost of the unconstrained polynomial root-finding step: for each candidate frequency the optimal $\theta_1 \geq 0$ solution is obtained by re-evaluating the periodogram at the *same* set of polynomial roots after restricting to those that produce a non-negative amplitude, with no additional root-finding required. The implementation accompanying this paper includes this constant-time filter. The multiband solver (Appendix B) applies it by default, falling back to the unconstrained maximum only in the degenerate case where *no* candidate root yields $\theta_1 \geq 0$; the single-band `power(...)` retains the unconstrained behavior for continuity with Lomb–Scargle conventions and exposes the sign of the best fit, and we caution the user to check that the best-fit θ_1 is positive when using it.

2.2. *Extending to multi-band observations*

2.2.1. *Multi-phase model*

As shown in VanderPlas & Ivezić (2015), the multiphase periodogram (their $(N_{\rm base}, N_{\rm band}) = (0, 1)$ periodogram), for any model can be expressed as a linear combination of single-band periodograms:

$$P^{(0,1)}(\omega) = \frac{\sum_{k=1}^{K} V_{0,k}\, P_k(\omega)}{\sum_{k=1}^{K} V_{0,k}} \tag{28}$$

where K denotes the number of bands, $V_{0,k}$ is the weighted sum of squared residuals between the data in the k-th band and its weighted mean $\langle y \rangle$, and $P_k(\omega)$ is the periodogram value of the k-th band at the trial frequency ω.

With Equation 28, the template periodogram is readily applicable to multi-band time series, which is crucial for experiments like LSST, SDSS, Pan-STARRS, and other current and future photometric surveys.

Other multi-band extensions of the template periodogram are provided in Appendix B.

2.3. *Computational requirements*

For a given number of harmonics H, assembling the stationarity polynomial of Equation 22—whose degree is at most $6H - 2$—from the trigonometric sums requires $\mathcal{O}(H^2)$ operations. The reference Python implementation (`method='eigvals'`) locates the optimum by rooting this polynomial via its companion-matrix eigenvalues (§2), at a cost of $\mathcal{O}(H^3)$ per trial frequency. The default maximiser of the accompanying implementation (`method='scan'`) instead evaluates the phase objective directly on $\mathcal{O}(H)$ trial phases and Newton-polishes the bracketed maxima, so that the coefficient *assembly*—not root-finding—sets the per-frequency cost at $\mathcal{O}(H^2)$; only at the frequencies where the template-variance polynomial $|MM|$ nearly vanishes on the unit circle (rank-deficient or phase-clustered sampling) does it fall back to the exact $\mathcal{O}(H^3)$ root path and keep the better of the two solutions, so the $\mathcal{O}(H^2)$ saving is opportunistic rather than guaranteed (Appendix B develops the same construction for the multi-band modes). The realised per-frequency scaling therefore interpolates between $\mathcal{O}(H^2)$—the regime that holds throughout the practical harmonic range $H \lesssim 15$, where the fallback rarely fires—and the reference $\mathcal{O}(H^3)$ as the fallback incidence grows (at very high H, or on sparse or otherwise deep-$|MM|$ sampling).

When considering N_f trial frequencies, the per-frequency maximisation therefore scales as $\mathcal{O}(H^2 N_f)$ on the default path ($\mathcal{O}(H^3 N_f)$ on the reference path), while the computation of the sums (Equations 23 – 27) scales as $\mathcal{O}(HN_f \log HN_f)$.

Therefore, on its default scan path, the entire template periodogram scales as

$$\mathcal{O}(HN_f \log HN_f) \;+\; \mathcal{O}(H^2 N_f), \tag{29}$$

with the reference root-finding path replacing the second term by $\mathcal{O}(H^3 N_f)$.

We have written the bound as the sum of two independent contributions in order to emphasize that the two terms are governed by different operations and dominate in different regimes. The first term, from the non-equispaced fast Fourier transforms used to compute the C and S sums of Equations 23–27, dominates at low H and large N_f – the regime typical of large surveys with near-sinusoidal templates. The second term, from assembling and maximising over the degree-($\leq 6H-2$) polynomial of Equation 22 at every trial frequency, dominates at moderate-to-high H and moderate N_f – the regime typical of sharply peaked templates such as eclipsing-binary or planet-transit shapes.

However, an important consideration is that the peak width scales inversely with the number of harmonics $\delta f_{\rm peak} \propto 1/H$ and so the number of trial frequencies needed to resolve a peak increases linearly with the number of harmonics in the template.

The computational scaling factoring in an extra power of H is therefore

$$\mathcal{O}(H^2 N_f \log HN_f + H^3 N_f) \tag{30}$$

on the default path (again with $H^3 N_f \to H^4 N_f$ on the reference root-finding path).

For a fixed number of harmonics H, the template periodogram scales as $\mathcal{O}(N_f \log N_f)$. However, for a constant number of trial frequencies N_f, the per-frequency cost grows as $\mathcal{O}(H^2)$ on the default scan path ($\mathcal{O}(H^3)$ on the reference root-finding path), and computational resources alone limit H to reasonably small numbers $H \lesssim 15$ (see Figure 1).

This practical ceiling has a direct bearing on the algorithm's scope of applicability for box-like signals such as exoplanetary transits and detached eclipsing binaries. As we calibrate in Section 2.4, the canonical analytic trapezoidal-transit and detached-EB templates considered here have slope (first-derivative) discontinuities at their ingress/egress corners, which produce a $\sim 1/n^2$ tail in the Fourier coefficients and require $H \approx 12$ to reach 95% L^2 fidelity and $H \approx 21$ to reach 99% (Table 1). FTP therefore *can* handle transit- and eclipse-shaped signals, but at the upper end of computationally tractable H values, and at a cost that grows as $H^2 N_f$ on the default path ($H^3 N_f$ on the reference). Real limb-darkened transits are smoother than the trapezoidal idealisation—they have finite, gently-curved ingress/egress and consequently Fourier coefficients that decay faster than the trapezoid's $1/n^2$ tail—so the H requirements in practice may be lower than Table 1 suggests for a pathological trapezoid. Nevertheless, for surveys targeting genuinely box-shaped signals with very sharp ingress/egress (e.g. short-cadence shallow transits or short-duration eclipses), transit-tailored methods such as the Box Least-Squares periodogram (BLS, Kovács et al. 2002)—of which a GPU-accelerated descendant, CETRA, has recently been described by Smith et al. (2025)—are likely to be a more efficient choice than FTP.

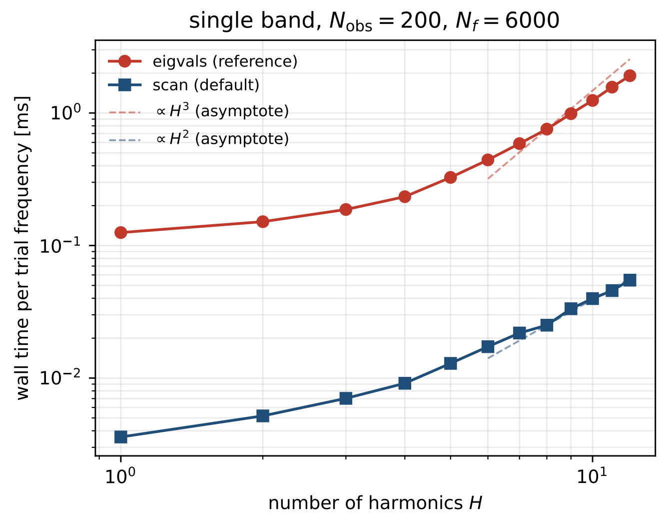

FIG. 1.— Wall time per trial frequency versus the number of harmonics H for the default scan+polish maximiser (`method='scan'`, blue) and the reference polynomial-root implementation (`method='eigvals'`, red), timed on identical well-conditioned single-band data ($N_{\rm obs} = 200$, $N_f = 6000$). The default path is a measured factor of 25–35 faster across all H. At low H both paths are dominated by the shared $\mathcal{O}(HN_f \log HN_f)$ NFFT summations and rise only mildly with H; by $H \gtrsim 8$ the per-frequency maximiser takes over, and the reference cost steepens toward its $\mathcal{O}(H^3)$ root-finding asymptote while the scan tracks $\mathcal{O}(H^2)$ (dashed segments; the measured $8 \to 12$ log–log slopes are $\approx$2.3 and $\approx$1.9 respectively).

2.4. *Choosing the harmonic order H*

The harmonic order H is a free parameter of the algorithm. Two considerations bound it in practice: (i) the L^2 fidelity of the H-truncated Fourier series as an approximation of the true template shape, and (ii) the numerical precision of the polynomial-rooting step at large H.

To quantify (i) for representative fixed-shape signals, we evaluated the cumulative explained-variance fraction $F(H) = \sum_{n=1}^{H}(c_n^2 + s_n^2)/\sum_{n=1}^{\infty}(c_n^2 + s_n^2)$ for five canonical analytic templates: an asymmetric exponential RR Lyrae ab pulse (fast rise, slow decline); a skewed sinusoidal RR Lyrae c profile; a trapezoidal planet transit with linear ingress/egress; a detached eclipsing binary built from two such trapezoidal eclipses; and a Doppler-beaming + ellipsoidal-modulation profile (exactly two-harmonic by construction). Table 1 lists the smallest H that reaches each of the $F = 0.90, 0.95, 0.99$ thresholds, and Figure 2(a) shows the full $F(H)$ curves.

The five templates span an order of magnitude in their harmonic requirements. At the low end, the ellipsoidal + beaming profile is exactly band-limited at $H = 2$; the skewed-sinusoid RR Lyrae c reaches 99% at $H = 3$. The asymmetric-pulse RR Lyrae ab profile reaches 99% at $H = 8$. At the high end, the trapezoidal transit and detached-EB templates have slope (first-derivative) discontinuities at their ingress/egress corners, which gives a $\sim 1/n^2$ tail in the Fourier coefficients; both require $H \approx 12$ for 95% fidelity and $H \approx 21$

TABLE 1
THE MINIMUM HARMONIC ORDER $H_{\rm min}(F)$ AT WHICH THE H-TRUNCATED FOURIER EXPANSION OF EACH CANONICAL TEMPLATE CAPTURES THE LISTED FRACTION F OF ITS TOTAL L^2 ENERGY. THE RIGHT-HAND COLUMNS REPORT THE EMPIRICAL PERIOD-RECOVERY RATE AT PEAK-TO-PEAK SNR$=7$ ON 30 INDEPENDENT SYNTHETIC LIGHT CURVES WITH $N_{\rm obs}=100$ IRREGULAR SAMPLES PER TRIAL, COMPARING THE FAST TEMPLATE PERIODOGRAM AT $H=H_{\rm min}(F{=}0.95)$ WITH THE SINGLE-HARMONIC LOMB–SCARGLE-EQUIVALENT (FTP AT $H{=}1$). TABULATED VALUES COME FROM THE ANALYTIC FOURIER EXPANSIONS DESCRIBED IN SECTION 2.4 AND FROM THE OPEN-SOURCE REPRODUCTION SCRIPT IN THE PAPER REPOSITORY.

Template	$H_{\rm min}(F)$			Recovery at SNR$=7$		
	90%	95%	99%	H	FTP	LS-eq.
RR Lyrae ab	2	3	8	$H=3$	100%	100%
RR Lyrae c	2	2	3	$H=2$	100%	100%
Planet transit	6	12	21	$H=12$	100%	47%
Detached EB	6	12	21	$H=12$	100%	0%
Ellipsoidal + beaming	2	2	2	$H=2$	100%	0%

for 99%. These tabulated values are specific to the 12 per cent duty cycle of the analytic trapezoids used here: for a pulse of duty cycle δ the Fourier energy extends to harmonics $n\sim 1/\delta$, so the required harmonic order scales as $H_{\rm min}(F)\propto 1/\delta$—a 6 per cent duty-cycle transit needs roughly twice the tabulated H. For applications where the underlying signal has sharper features than these analytic templates (e.g. true limb-darkened transits with fast cadence), the 99% values may need to be revised upward.

The practical consequence for period detection is shown in Figure 2(b), which reports the period-recovery rate of FTP and of the LS-equivalent (FTP at $H{=}1$, i.e. a single-harmonic sinusoidal template) on synthetic noisy light curves. Each light curve consists of $N_{\rm obs}=100$ samples drawn uniformly from a $T_{\rm obs}=100\,$day window with independent Gaussian noise scaled to the indicated peak-to-peak SNR; each periodogram is evaluated on the frequency window $[0.01,5]$ cyc day^{-1} at 10 samples per peak, and the period is declared recovered when $|f_{\rm peak}-f_{\rm true}|/f_{\rm true}<5\times10^{-3}$. Thirty independent trials are run per (template, SNR) cell. For the smooth, near-sinusoidal RR Lyrae profiles the two estimators agree at all SNRs: at this dense sampling ($N_{\rm obs}=100$), a single-harmonic template suffices. (This conclusion is specific to the dense-sampling regime; in the sparse regime, where few epochs constrain many free parameters, the template's shape prior becomes decisive even for these smooth profiles, as quantified in the sparse-sampling simulation-validation results of Section 5.) For the three non-sinusoidal templates (transit, EB, ellipsoidal + beaming), FTP at $H=H_{95\%}$ recovers the period in $\geq 29/30$ trials at every SNR tested, while the LS-equivalent fails completely for EB and ellipsoidal + beaming (0/30 recoveries at all SNRs) and only partially for the transit template (4/30, 14/30, and 17/30 at SNR$=$ 3, 7, 15 respectively). This is the regime in which the H-truncation matters: harmonics beyond the first are required to match the underlying signal shape.

For the numerical-precision consideration (ii), we have measured the maximum periodogram error of the implementation in this paper against a brute-force evaluation that explicitly maximises $V(\theta,\omega)$ on a dense θ_2 grid with local refinement. Verified through $H=21$ (covering the highest harmonic orders required by Table 1), the maximum absolute error in $P(\omega)$ is $\sim10^{-15}$ with direct summations and $\sim10^{-13}$ with the NFFT-based summations.[6] The guarantee is one-sided by construction: every candidate root is projected onto the unit circle and the *true* periodogram is evaluated there, so the reported power – a maximum over a finite candidate set – can only *under*estimate the true per-frequency maximum, never overestimate it. These floors are measured under homoscedastic measurement errors. They degrade when a single observation dominates the total inverse-variance weight budget by a factor $\gtrsim 10^8$ – catastrophic cancellation in the weighted moment sums, which afflicts the fast NFFT-based summations earlier than direct summations – and this condition is the practical meaning of the "pathological cases" caveat in the abstract. The floor is well below any astrophysically relevant amplitude threshold, so polynomial-rooting error does not limit FTP at the values of H considered in Table 1.

3. IMPLEMENTATION

An open-source implementation of the Fast Template Periodogram (FTP) in Python is available.[7] Polynomial algebra is performed using the `numpy.polynomial` module (Jones et al. 2001–). The `nfft` Python module, [8] which provides a Python implementation of the non-equispaced fast Fourier transform, is used to compute the necessary sums for a particular time series.

The high-level entry point, `autopower(samples_per_peak=...)`, exposes the trial-frequency grid density as a user parameter: `samples_per_peak` is the number of trial frequencies that cover one full-width of a periodogram peak. Because the peak width scales roughly as $\delta f_{\rm peak}\propto 1/H$ (Section 2.3), the value of `samples_per_peak` must scale at least linearly with H to avoid under-sampling the peaks. In particular, the shipped default `samples_per_peak` $=5$ is a reasonable choice only for near-sinusoidal ($H=1$) templates: it implies under one trial frequency per peak at $H=10$, and even a denser `samples_per_peak` $=10$ grid – the conventional Lomb–Scargle choice – leaves peaks only about three samples wide in an empirical inspection at $H=7$, so on the default grid the true frequency can be missed in favor of a noise peak. For the sharper templates studied in Section 2.4 – in particular the trapezoidal-transit and detached-EB shapes, which require $H_{95\%}=12$ in Table 1 – `samples_per_peak` $\gtrsim 30$ is a more defensible default. This is the runtime/grid-density versus peak-recovery tradeoff documented in the open-source implementation, and users selecting H via Section 2.4 should adjust `samples_per_peak` accordingly.

No explicit parallelism is used anywhere in the current implementation, however certain linear algebra operations in `Scipy` use OpenMP via calls to BLAS libraries that have OpenMP enabled.

[6] Issue #33 of the FTP repository at https://github.com/PrincetonUniversity/FastTemplatePeriodogram/issues/33 documents the test methodology and the precision floor of `numpy.polynomial.polyroots` as used by the implementation.
[7] https://github.com/PrincetonUniversity/FastTemplatePeriodogram
[8] https://github.com/jakevdp/nfft

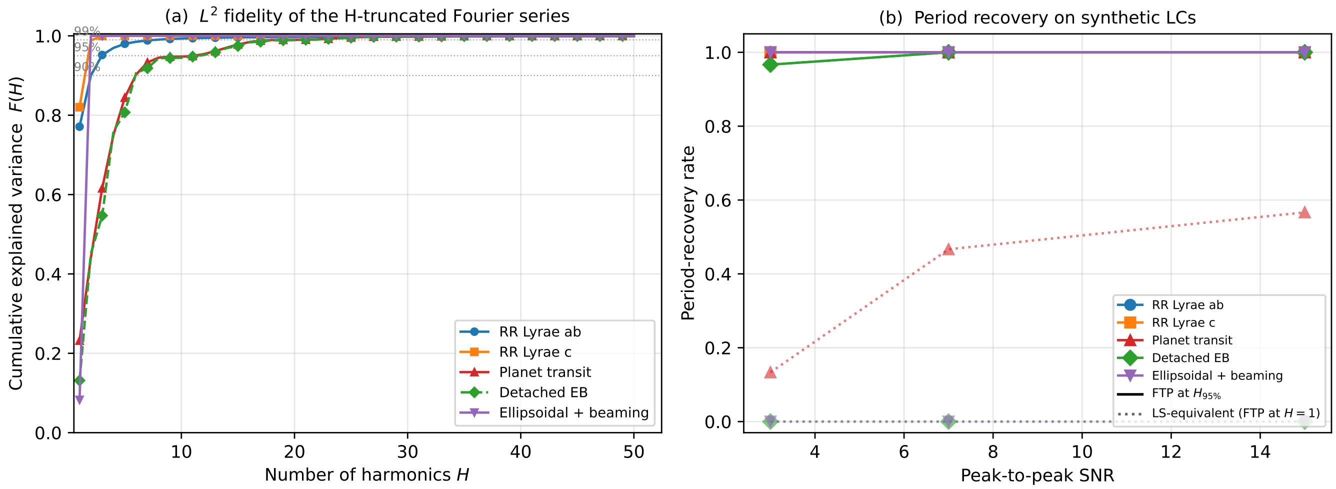

FIG. 2.— (a) Cumulative L^2 explained-variance fraction $F(H)$ of the H-truncated Fourier series as an approximation of each canonical template. Horizontal dotted guides mark $F = 0.90, 0.95, 0.99$. Detached EB and Planet transit have visually overlapping curves (both driven by sharp ingress/egress discontinuities); the dashed green line distinguishes the EB. (b) Empirical period-recovery rate as a function of peak-to-peak signal-to-noise ratio for densely-sampled $N_{\rm obs} = 100$ synthetic light curves (30 trials per cell), comparing FTP at $H = H_{95\%}$ (solid) with the Lomb–Scargle-equivalent FTP at H=1 (dotted) for the same five canonical templates. FTP at the per-template $H_{95\%}$ recovers the period in $\geq$ 96% of trials across the SNR range. The LS-equivalent agrees with FTP for the smooth RR Lyrae profiles (overlapping solid lines), fails completely for the EB and ellipsoidal + beaming templates (flat dotted lines at zero), and recovers only a fraction of trial transits (red dotted line, climbing from ∼13% at SNR = 3 to ∼57% at SNR = 15).

All timing tests were run on a quad-core 2.6 GHz Intel Core i7 MacBook Pro laptop (mid-2012 model) with 8GB of 1600 MHz DDR3 memory. The `Scipy` stack (version 0.18.1) was compiled with multi-threaded MKL libraries.

3.1. *Comparison with non-linear optimization*

In order to evaluate the accuracy and speed of the template periodogram, we have included slower alternative solvers within the Python implementation of the FTP that employ non-linear optimization to find the best fit parameters.

Figure 3 shows example periodograms computed for simulated data, and compares the FTP to the multi-harmonic Lomb–Scargle (MHLS; Bretthorst et al. 1988; Schwarzenberg-Czerny 1996; Palmer 2009) and Box Least Squares (BLS; Kovács et al. 2002) periodograms; the remaining accuracy and timing comparisons (Figures 4 and 5) use the same simulated-data prescription. The simulated data have uniformly random observation times, with Gaussian-random, homoskedastic, uncorrelated uncertainties. An eclipsing binary template, generated by fitting a well-sampled, high signal-to-noise eclipsing binary in the HATNet dataset (2MASS J02240150+5717241, also known as BD+56° 603) with a 10-harmonic truncated Fourier series, is added to the simulated lightcurves and used as the template for the FTP search. At the correct period, the FTP yields a much stronger detection than either MHLS—which has $2H{+}1$ free parameters and consequently reduces the signal-to-noise of the periodogram peak relative to a fixed-shape template—or BLS, which provides only a coarse box-shaped match to the smooth EB profile.

3.1.1. *Accuracy*

For weak signals or signals folded at the incorrect trial period, the V surface in θ_2 admits up to $6H{-}2$ stationary points with $YM \neq 0$ (the degree bound of the polynomial in Equation 22; up to $2H$ further stationary points where $YM = 0$ are zero-power minima, factored out of Equation 22), so non-linear optimization algorithms can fail to find the global minimum and instead lock onto one of the spurious local extrema. The FTP sidesteps this entirely by enumerating all roots of Equation 22 and choosing the one that yields the largest $P(\omega)$, so it recovers the optimal solution even when the signal is weak or absent.

Figure 4 illustrates the accuracy improvement with FTP. Many solutions found via non-linear optimization are significantly suboptimal compared to the solutions found by the FTP.

Figure 5 compares FTP results obtained using the full template ($H = 10$) with those obtained using smaller numbers of harmonics. The left-most plot compares the $H = 1$ case (weighted Lomb-Scargle), which, as also demonstrated in Figure 3, illustrates the advantage of the template periodogram for known, non-sinusoidal signal shapes.

3.1.2. *Computation time*

On its default scan path the FTP scales asymptotically as $\mathcal{O}(N_f H \log N_f H)$ with respect to the number of trial frequencies, N_f, and as $\mathcal{O}(N_f H^2)$ with respect to the number of harmonics in which the template is expanded, H (the reference root-finding path replaces $N_f H^2$ by $N_f H^3$). For a given resolving power, however, there is an additional factor of H in each of these terms due to the number of trial frequencies necessary to resolve a periodogram peak being proportional to H. For all reasonable N_f the computation time is dominated by computing the polynomial coefficients and the per-frequency phase maximisation, both of which scale linearly in N_f.

The number of trial frequencies needed for finding astrophysical signals in a typical photometric time series is

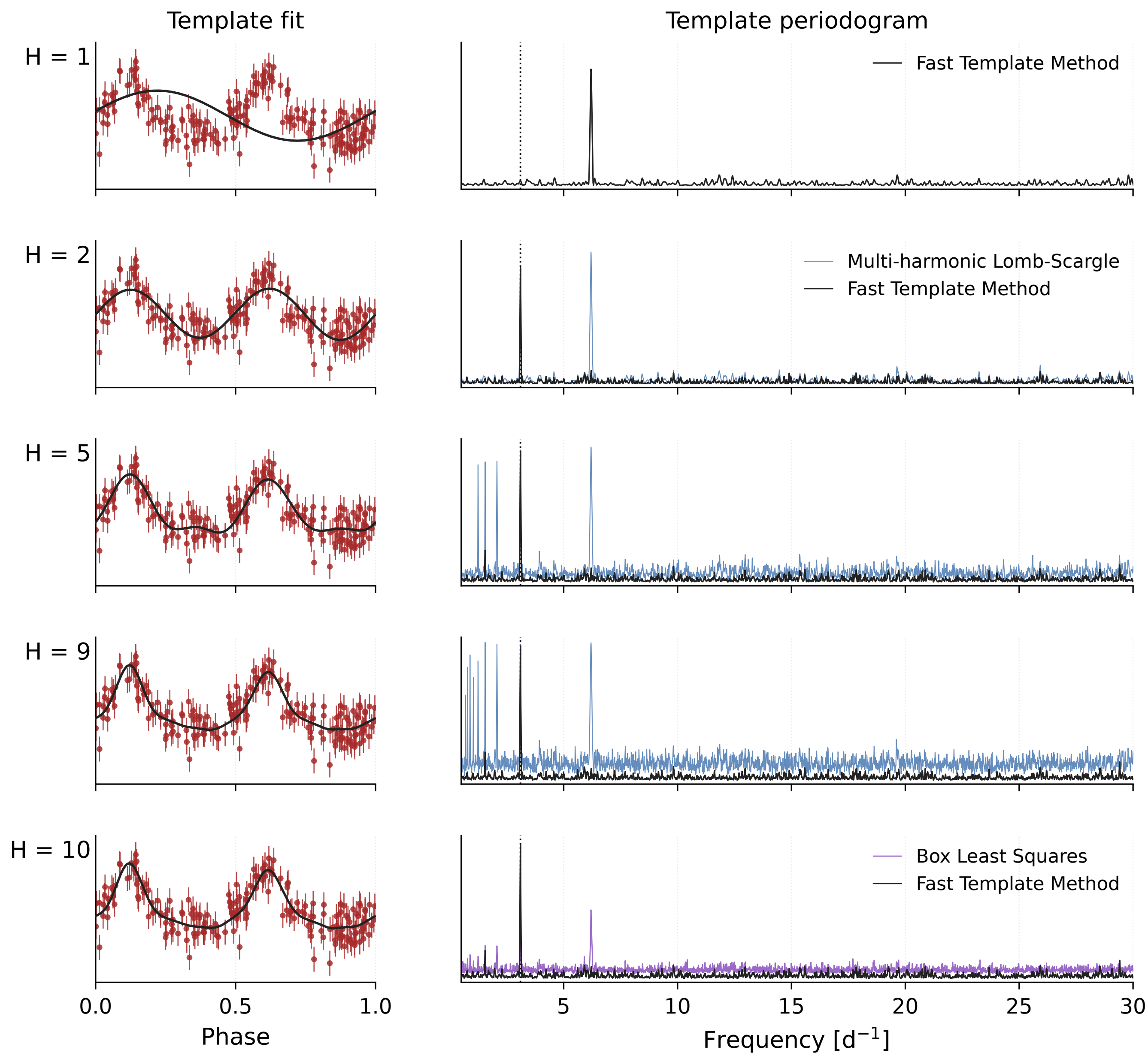

FIG. 3.— Template periodograms performed on a simulated eclipsing binary lightcurve (shown phase-folded in the left-hand plots). The top-most plot uses only one harmonic, equivalent to a Lomb-Scargle periodogram. Because the simulated eclipsing binary has two minima per orbital cycle, this single-harmonic periodogram peaks at twice the true frequency; two or more harmonics are required to recover the correct period. Subsequent plots use an increasing number of harmonics, which produces a narrower and higher peak height around the correct frequency. For comparison, the multi-harmonic extension to Lomb-Scargle is plotted in blue, using the same number of harmonics as the FTP. The Box Least-Squares (Kovács et al. 2002) periodogram is shown in the final plot.

$$N_f = 1.75 \times 10^6 \left(\frac{H}{1}\right)\left(\frac{\text{baseline}}{10\text{ yrs}}\right)\left(\frac{\alpha}{5}\right)\left(\frac{15\text{ mins}}{P_{\text{min}}}\right) \tag{31}$$

where α represents the "oversampling factor," $\Delta f_{\text{peak}}/\Delta f$, where $\Delta f_{\text{peak}} \sim 1/\text{baseline}$ is the typical width of a peak in the periodogram and Δf is the frequency spacing of the periodogram.

The per-frequency phase maximisation dominates the run time in this regime, which is precisely why replacing the reference polynomial root-finding with the scan+polish maximiser—now the default in the accompanying implementation—is worthwhile. On well-conditioned single-band data the default path is a measured factor of 25–35 faster than the reference Python implementation (`method='eigvals'`), and this factor is grid-stable in N_f: at $H = 8$ we measure $\approx$32×, $\approx$29× and $\approx$28× at $N_f = 2{,}000$, 8,000 and 20,000 respectively (Figure 1).

Figure 6 compares the timing of the FTP (default scan path) with that of previous methods that employ non-linear optimization at each trial frequency. Over the range where the non-linear solver remains tractable, the default FTP is already a measured factor of $\sim$14–15 faster at the sparsest, most rank-deficient test case (15

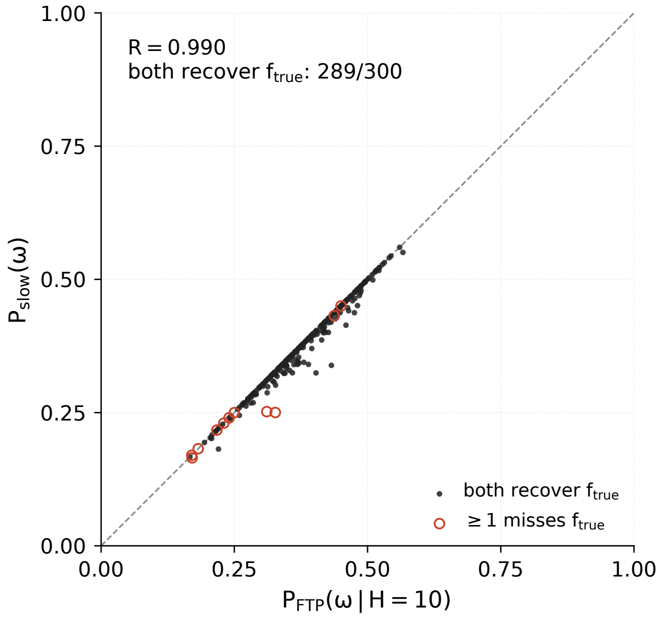

Fig. 4.— Comparing accuracy between previous methods that rely on non-linear optimization at each trial frequency with the fast template periodogram described in this paper. Each point corresponds to one realization of the simulated data (an example of which is shown in Figure 3); the axes give the *peak* periodogram value $P_{\rm FTP}(\omega\,;\,H{=}10)$ from the FTP and $P_{\rm slow}(\omega)$ from the non-linear "slow" template fitter, not the periodograms themselves. The annotated R is the Pearson linear-correlation coefficient between the two sets of peak values. The FTP consistently finds more optimal template fits than the non-linear solver, which does not guarantee convergence to a globally optimal solution; in realizations where the slow fitter is trapped in a local V-minimum, it reports a higher residual V and therefore a lower peak $P_{\rm slow}$ than the FTP, producing the points below the diagonal at low $P_{\rm FTP}$.

datapoints, where the scan itself falls back to the exact root path) and ∼480–500 faster by $N_{\rm obs} = 250$, in both the constant-cadence ($N_f \propto N_{\rm obs}$) and constant-baseline (N_f fixed) regimes. Because non-linear optimization scales as $\mathcal{O}(N_f N_{\rm obs})$ while the FTP's per-frequency cost is independent of $N_{\rm obs}$, this advantage grows without bound as $N_{\rm obs}$ increases (dotted extrapolation in Figure 6).

3.2. *Significance and false-alarm calibration*

At a fixed trial frequency the template periodogram is not the score of a linear model of fixed dimension: the model $\hat{y} = \theta_1 \mathbf{M}_{\theta_2} + \theta_3$ has only three free parameters, but the phase θ_2 enters non-linearly, and the reported power is a maximum over the up-to-$(6H{-}2)$ stationary phases of Equation 22. The classical per-frequency null for linear periodograms (an incomplete-beta / χ^2 law with the model's degrees of freedom) therefore does not apply directly; the natural framework is the extreme-value treatment of Baluev (2008), with an *effective* number of degrees of freedom inflated relative to the single-sinusoid case. Rather than derive a bound, we measure the null directly: 3000 pure-noise light curves at a production-like cadence (single band, $N_{\rm obs} = 40$, three-year baseline with 270-day observing seasons, homoscedastic $\sigma \approx 0.010\,$mag), searched over 10^4 trial frequencies on $[1,5]\,$cyc day^{-1} with RR Lyrae-medoid templates truncated at order H, alongside a floating-mean (generalised) Lomb–Scargle baseline (GLS; Zechmeister & Kürster 2009). Two facts summarise the measurement (Figure 7, left). First, at $H = 1$ the FTP *is* the GLS: the per-realization maximum powers agree to a maximum absolute difference of 1.2×10^{-9} over all 3000 realizations, as they must – a single-harmonic template spans the same linear space as a floating-mean sinusoid. Second, the inflation from the non-linear phase family is modest and saturates almost immediately: the mean null maximum rises from 0.407 at $H = 1$ to 0.440 at $H = 3$, then 0.442 ($H = 6$) and 0.443 ($H = 12$), and the empirical FAP = 0.01 threshold rises from 0.535 ($H = 1$) to 0.551, 0.551, and 0.553 at $H = 3$, 6, and 12. An $H = 12$ template therefore costs less than +0.02 in detection threshold at the calibrated FAP = 0.01 relative to Lomb–Scargle – far below the cost of the $2H{+}1 = 25$ free coefficients of a multi-harmonic fit of the same order – because the fixed template shape confines the maximisation to a one-parameter curve of phase shifts inside the $(2H{+}1)$-dimensional harmonic space. Generalised-extreme-value fits (dashed curves in Figure 7) describe the empirical curves well and extend the thresholds beyond the Monte-Carlo resolution.

The statistic being calibrated is the normalised power of Equation 2, $P = (V_0 - V)/V_0$ – equivalently $1 - \chi^2/\chi^2_0$, since $\chi^2 = WV$ (Section 1.1) – which is confined to $[0,1]$: $V \geq 0$ gives the upper bound, and the constant-only null model is nested in the template model, so $V \leq V_0$. In the multi-band hierarchy of Appendix B the reference variance V_0 differs by sharing mode (Table 3): the three modes that retain a per-band offset (`independent`, `shared_phase`, `floating_offsets`) measure power against the per-band-centred variance $\sum_k W^{(k)} YY^{(k)}$, whose null model is a separate constant per band, whereas the fully shared `sesar` mode measures power against the global weighted variance of the data after subtracting its fixed per-band offsets $\lambda^{(k)}$, whose null model is a single global constant. Null distributions, and therefore detection thresholds, calibrated for one mode do not transfer to another, even on identical data.

When a vocabulary of K templates is searched and the maximum power over the vocabulary is reported (the catalog-max statistic $\max_{j \leq K} \max_\omega P_j(\omega)$), the null acquires a K-dependence: an extra-trials cost. A Bonferroni bound, ${\rm FAP}_K(z) \leq K\,{\rm FAP}_1(z)$, is always valid but conservative, because the per-template statistics are computed on the same data and the templates share their low-order harmonics. The measured null (Figure 7, right; nested prefixes of a single $K{=}8$ partitioning-around-medoids vocabulary built from the Sesar et al. 2010 RR Lyrae library) quantifies both statements: the mean null maximum grows from 0.422 ($K{=}1$) to 0.467 ($K{=}8$) and the FAP = 0.01 threshold from 0.540 to 0.576, while at that measured $K{=}8$ threshold a single template has FAP = 0.0030, so the Bonferroni estimate ($8 \times 0.0030 = 0.024$) overstates the measured 0.010 by a factor 2.4. This extra-trials cost is a shift of the null in power units and is not directly commensurate with a recovery *rate*; the appropriate yardstick for the small, non-monotonic wobble of period-recovery rate with vocabulary size in the sparse-sampling recovery simulations (Section 5.3; RRab-dominated universe, 768 pooled sources per cell) is the binomial sampling uncertainty: the pooled rate at

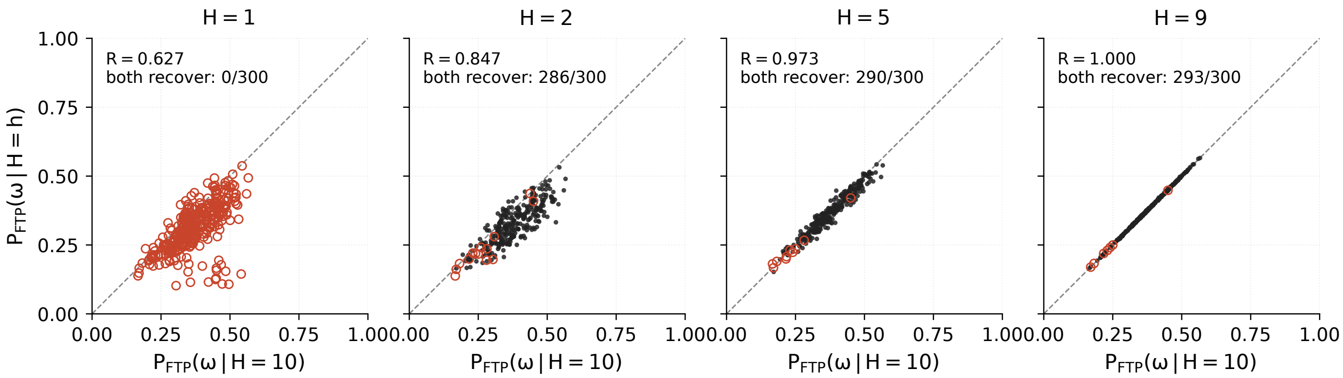

FIG. 5.— Similar to Figure 4. Here we compare the template periodogram calculated with $H = 10$ harmonics to the template periodogram using a smaller number of harmonics $H < 10$. Each point is the *peak* periodogram value from one realization; R is again the Pearson linear-correlation coefficient between the two sets of peak values. The template and data prescription match those of Figure 3.

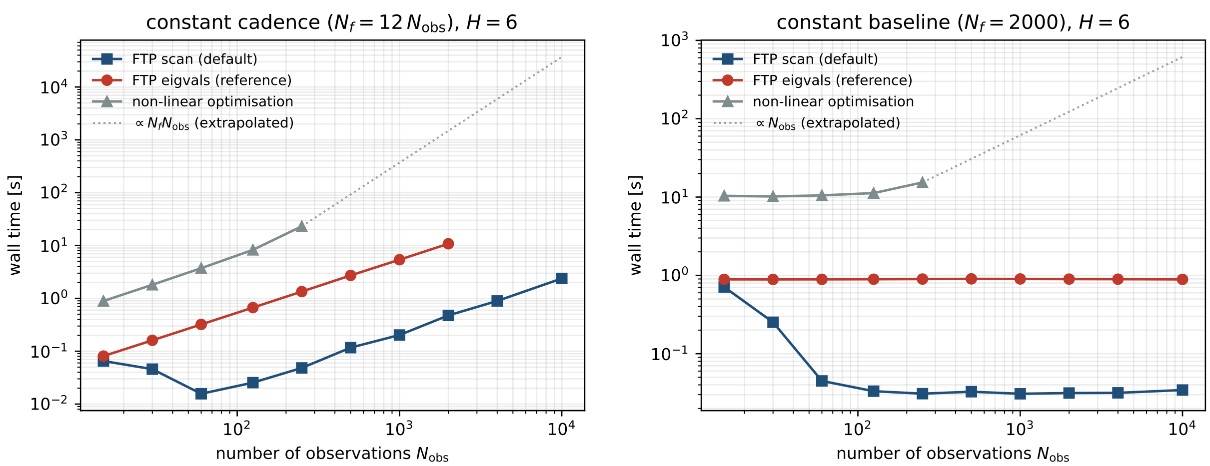

FIG. 6.— Computation time of the FTP (default scan path, `method='scan'`, blue; reference root-finding path, `method='eigvals'`, red) compared with full non-linear optimization at each trial frequency (grey; 10 random phase restarts per frequency). *Left*: timing for the case when $N_f = 12N_{\rm obs}$, i.e. the cadence of the observations is constant. *Right*: timing for the case when N_f is fixed, i.e. the baseline of the observations is constant. Non-linear optimization scales as $\mathcal{O}(N_f N_{\rm obs})$ (dotted extrapolation beyond the largest tractable point) while the FTP scales as $\mathcal{O}(HN_f \log HN_f + N_f H^2)$ on its default path ($N_f H^2 \to N_f H^3$ on the reference path), independent of $N_{\rm obs}$; H is the number of harmonics needed to approximate the template. In the constant-baseline panel the scan cost rises toward the reference at the sparsest $N_{\rm obs}$, where the rank-deficient sampling triggers the exact-root fallback.

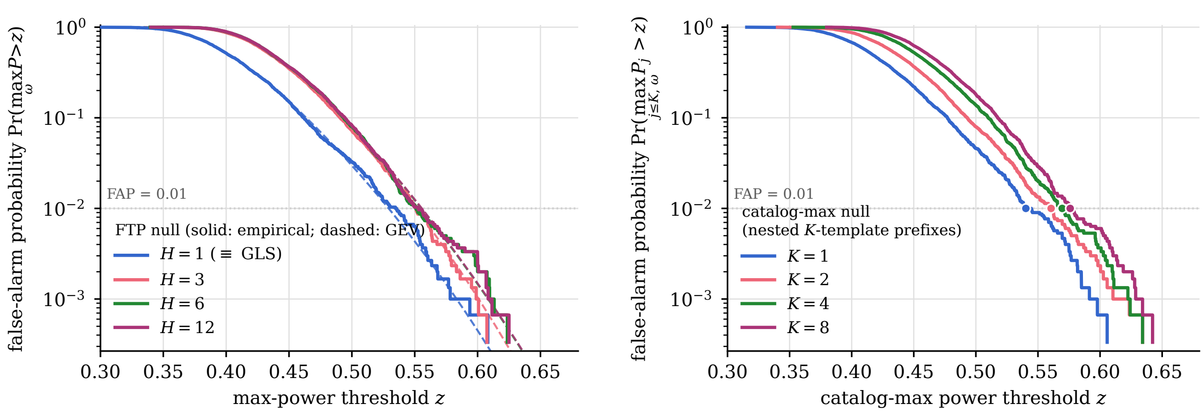

FIG. 7.— Measured pure-noise null of the maximum periodogram power (3000 realizations; single band, $N_{\rm obs} = 40$, three-year seasonal cadence, homoscedastic $\sigma \approx 0.010\,$mag; 10^4 trial frequencies on $[1, 5]\,$cyc day^{-1}). *Left*: false-alarm probability versus max-power threshold for the FTP at $H \in \{1, 3, 6, 12\}$ (solid: empirical survival function; dashed: generalised-extreme-value fit). The H=1 curve is identical to GLS; the inflation from the non-linear phase family appears at $H \geq 3$ and is nearly saturated there. *Right*: null of the catalog-max statistic for nested K-template prefixes of a single K=8 vocabulary; markers show the empirical FAP = 0.01 thresholds, which grow from 0.540 (K=1) to 0.576 (K=8) – the pure extra-trials cost of a larger vocabulary.

$K = 1, 2, 4, 8$ is 0.716, 0.715, 0.738, 0.741, with Wilson 95% intervals $[0.683, 0.747]$, $[0.682, 0.746]$, $[0.706, 0.768]$, $[0.709, 0.771]$ – a non-monotone range of 0.026 with mutually overlapping intervals, and therefore not evidence of a vocabulary-quality effect at this sampling.

Two caveats bound this calibration. First, the null is calibrated at a single configuration – one band, $N_{\rm obs} = 40$, this cadence and grid – and the thresholds shift with the number of observations, the band structure, and the grid density, so they do not transfer to other survey configurations without recalibration. Second, the null's K-axis isolates the pure extra-trials cost by nesting prefixes of one fixed K=8 vocabulary, whereas the production K-sweep rebuilds an independent vocabulary at each K – the template identities themselves change with K – an effect bounded by the measured $\sim$0.025 spread of the per-template null-max means across the eight vocabulary templates. Finally, this section calibrates *significance* only: the completeness and purity of template-based detection at a controlled false-alarm rate are survey-specific questions deferred to follow-up work.

4. TEMPLATE VOCABULARIES FOR HETEROGENEOUS SIGNAL CLASSES

The preceding sections assume a single, *a priori* known template. Real signal classes are heterogeneous – RR Lyrae alone span fundamental-mode (ab) and first-overtone (c) morphologies – so a blind search must instead carry a small *vocabulary* of K templates and report the per-frequency maximum power over the set (the catalog-max statistic whose null is calibrated in Section 3.2). The computational cost is K times the single-template cost, so the vocabulary must be small and representative. This section describes how the accompanying implementation constructs such vocabularies from a library of empirical light-curve shapes; Section 5 validates the resulting searches on simulated sparse multi-band surveys.

Shape space and invariances. Every library shape is stored as a truncated Fourier series (Equation 14) with no constant term, rescaled at construction to unit Fourier energy, $\sum_{k=1}^{H}(c_k^2 + s_k^2) = 1$. Vertical offset and overall scale – the parameters θ_3 and θ_1 that the periodogram itself fits per source – are therefore quotiented out before any shape comparison. Phase is removed by the dissimilarity itself, the orbit-minimized L^2 distance

$$d^2(f, g) \;=\; \min_{\Delta\in[0,1)} \left\| f - g(\cdot - \Delta) \right\|^2 \;=\; 1 - \max_{\Delta} \mathrm{CC}(\Delta), \tag{32}$$

where $\mathrm{CC}(\Delta)$ is the circular cross-correlation of the two harmonic-coefficient sequences ($w_k = z_k^{(f)}\overline{z_k^{(g)}}$ with $z_k = c_k - is_k$) and the norm is scaled so that a unit-energy shape has $\|f\| = 1$. The phase maximization is solved exactly per pair: CC is a degree-H trigonometric polynomial, evaluated on an alias-free oversampled FFT grid (size $\geq 8H+1$, rounded up to a power of two), after which every cyclic local maximum is polished by Newton iterations on the closed-form $\mathrm{CC}'/\mathrm{CC}''$; the reported distance is machine-precision, not grid-limited. Reflections (time-reversals) are deliberately *not* identified: the fast-rise/slow-decline asymmetry of an RRab light curve is a physical, discriminating feature, so a mirrored shape is a distinct template.

Unsupervised selection (PAM k-medoids). Given the symmetric matrix of pairwise orbit distances, the default selector is classical partitioning around medoids (Kaufman & Rousseeuw 1990) on the objective $\sum_i \min_m d(f_i, f_m)$ (unsquared distances): a greedy BUILD seeding followed by SWAP descent, which applies the single best strictly cost-reducing (medoid, non-medoid) exchange until none remains, with ten restarts (the deterministic BUILD plus nine seeded random size-K subsets) and the lowest-cost solution kept. Medoids are actual library members, so every vocabulary template is a physical light-curve shape rather than a cluster average. The vocabulary size K is an input, chosen externally by the recovery-versus-K experiments of Section 5. A fixed-chart alternative distance built on the classical Fourier invariants $R_{k1} = A_k/A_1$ and $\phi_{k1} = \phi_k - k\phi_1$ (Simon & Lee 1981) is retained purely as a validation diagnostic (agreement of nearest-medoid assignments with the orbit metric); it degenerates when the first-harmonic amplitude vanishes, so the orbit distance is the reference.

Recovery-driven selection (greedy set cover). The second selector optimizes the search objective directly. A seeded population of synthetic multi-band light curves is simulated once from the truth shapes and target cadence (Section 5) and frozen; for every library template the *standalone* mask of sources whose period that single template recovers is precomputed; and templates are then forward-selected by set cover – each step adds the template that recovers the most not-yet-recovered sources, with ties broken deterministically – stopping at K templates, at zero marginal gain, or at full coverage. Because the catalog statistic is a per-frequency maximum over the set, the union of standalone masks is used only as the selection signal; the returned figure of merit re-scores the chosen K-template catalog end-to-end. If coverage saturates before K selections, the set is padded to exactly K by farthest-point selection under the orbit distance (the saturation point is itself the empirical knee). An optional pre-filter restricts the candidate universe to the $M \geq K$ PAM medoids of the full library, capping the standalone-mask precompute; recovery-versus-K curves built on such a pool are lower bounds on the all-library curves. To keep the selection honest, the selection population is always simulated on a random split disjoint from any population used for evaluation.

Template libraries. Two empirical RR Lyrae libraries are used. The Sesar et al. (2010) SDSS Stripe 82 templates contribute one library shape per (star, band) pair; the g, r subset used here comprises 98 shapes, nearly all RRab. The Baeza-Villagra et al. (2025) DECam *griz* templates contribute 136 RRab and 144 RRc stars; their g, r subset comprises 560 shapes and is deliberately more morphologically diverse. Library curves are ingested by resampling each published template onto a uniform phase grid and taking the real FFT truncated at a common harmonic order ($H = 8$ throughout this section and the next), dropping the constant term and renormalizing to unit energy; libraries supplied at a different order are truncated or zero-padded to the common H and renormalized.

5. VALIDATION ON SIMULATED SPARSE MULTI-BAND SURVEYS

5.1. *Simulation harness and comparison methods*

Each experiment freezes a seeded population of 256 synthetic two-band (g, r) light curves per seed, with three seeds pooled for the headline cells (768 sources per cell). Every source draws a truth shape uniformly from the truth library, a period $P = 1/f$ with f uniform on the search band, and a random phase; the unit-energy shape is scaled by 0.5 mag (an rms amplitude of $0.5/\sqrt{2} \approx 0.35$ mag, shared across bands) around a mean magnitude of 15, and observed with heteroscedastic Gaussian noise $\sigma(m) = 0.01 + 0.005\, e^{m-20}$ (≈ 0.010 mag at the simulated magnitudes). Epochs are drawn independently per band, uniformly within 270-day yearly observing seasons across a three-year ($T = 1095.75$ d) baseline; a 60-epoch-per-band master cadence is downsampled to $N \in \{4, 8, 12, 16, 24, 40\}$ epochs per band, so all epoch counts score the *identical* sources and method contrasts are paired. A period is recovered when $|P_{\rm rec} - P_{\rm true}|/P_{\rm true} < 0.01$, with *no* credit for harmonic or window aliases (an alias-resolved re-scoring is given in Appendix C); uncertainties are Wilson 95% intervals on pooled counts and paired method contrasts are exact McNemar tests *on the same sources.* All methods reduce to $P_{\rm rec} = 1/{\rm argmax}$ over the identical explicit grid: 10^4 trial frequencies on $[1, 5]$ cyc day^{-1} (RR Lyrae periods 0.2–1 d), i.e. a grid spacing $\Delta f = 4.0\times10^{-4}$ cyc day^{-1} or 2.28 grid points per Rayleigh width $1/T$ – every run records Δf and this points-per-Rayleigh figure, and a hard guard rejects configurations with fewer than two points per Rayleigh width.

The comparison methods bracket the multi-band model space. The generalised Lomb–Scargle baseline (GLS; Zechmeister & Kürster 2009) fits one shared sinusoid plus per-band offsets – the $(N_{\rm base}, N_{\rm band}) = (1, 0)$ corner of the VanderPlas & Ivezić (2015) multi-band family – while the multi-band LS baseline (MBLS) gives each band its own floating-mean sinusoid sharing only the period, their $(0, 1)$ corner. The multi-harmonic Lomb–Scargle (MHLS; Schwarzenberg-Czerny 1996) is the free-shape reference: a shared Fourier series of requested order $H = 8$ plus per-band offsets, with the order capped per light curve at $H_{\rm eff} = \max\{1, \min[H, \lfloor (N_{\rm obs} - 2n_{\rm bands})/4 \rfloor]\}$ so that the design never reaches rank deficiency (an uncapped $H = 8$ design has $p \geq N_{\rm obs}$ columns at sparse N and interpolates every point, making the argmax meaningless); at 4 epochs per band the cap forces $H_{\rm eff} = 1$ and MHLS is verifiably identical to GLS. The conditional-entropy statistic (CE; Graham et al. 2013a) represents binned folding methods: each band min–max normalised, bands merged, and $H(m|\phi)$ minimised over a 10×5 (phase $\times$ magnitude) histogram. The FTP is run in catalog mode (`floating_offsets`, $H = 8$) with $K = 2$ vocabularies built by both selectors of Section 4 from the same library that generates the truth shapes (robustness to that assumption is tested below).

5.2. *Recovery versus epochs per band*

Figure 8 shows the headline result. At 8 epochs per band the PAM-vocabulary FTP recovers 0.902 [0.879, 0.921] of sources versus 0.417 [0.382, 0.452] for GLS – a paired contrast of 392 sources recovered only by the FTP against 19 only by GLS (McNemar $p \approx 10^{-91}$) – and 0.246 for MBLS, 0.089 for CE. At the sparsest

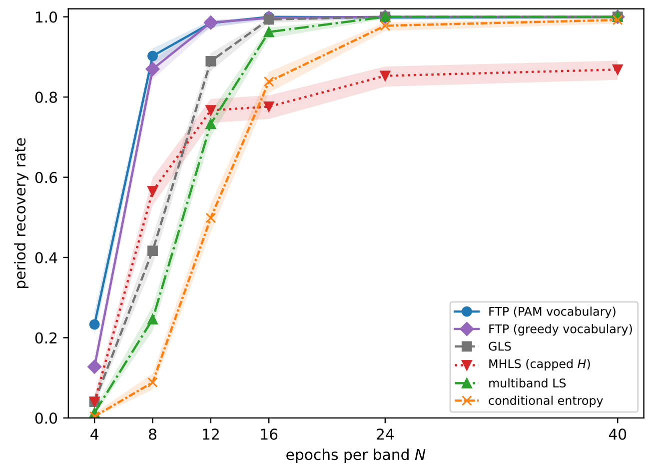

FIG. 8.— Period-recovery rate versus epochs per band on the RRab-dominated Sesar et al. (2010) universe (768 pooled sources per cell; bands are Wilson 95% intervals on pooled counts; vocabulary size $K = 2$). The template vocabulary dominates every free-shape and binned method in the sparse regime ($N \lesssim 12$ epochs per band) and converges with the sinusoidal baselines once the data constrain their models ($N \gtrsim 16$). MHLS plateaus below 0.9 because its extra freedom locks onto harmonic aliases (Appendix C). Absolute rates in the sparse cells are lower bounds at this grid density (see the grid-resolution caveat in the text).

cell ($N = 4$, where the MHLS cap makes every sinusoidal baseline identical to GLS) the FTP still recovers 0.233 [0.205, 0.264] versus 0.040 [0.029, 0.057] (169 vs. 21 discordant sources, $p \approx 10^{-29}$). By $N = 16$ the sinusoidal and template methods have all converged to $\gtrsim$0.96 (FTP vs. GLS: 5 vs. 0 discordant, $p = 0.06$), the binned CE statistic converges near $N = 40$ (80 merged points against 50 histogram cells), and MHLS alone plateaus at 0.868 [0.843, 0.891]: its $2H$ free coefficients happily fit the data at half or a third of the true frequency, and Appendix C shows its residual failures are indeed dominated by $2P/3P$ harmonic aliases rather than noise. The two vocabulary selectors agree closely on this shape-matched universe, with a small but significant PAM advantage in the sparsest cells (124 vs. 43 discordant at $N = 4$, $p \approx 3 \times 10^{-10}$); the greedy selector's value appears on the mismatched-library arm below.

Grid-resolution caveat. The absolute sparse-regime rates above are *lower bounds at the stated 10^4-frequency production grid* for the $H = 8$ and binned methods. A multi-harmonic peak has width $\sim$Rayleigh$/H$, so the production grid places only $\approx$0.29 points per $H = 8$ peak width. A paired convergence check re-scoring the identical seed-0 sources on a 2×10^4-frequency grid (4.56 points per Rayleigh width) finds 8 of 30 (method, N) cells significantly changed (exact McNemar, $p < 0.05$), *all* gaining at the denser grid: MHLS gains most (up to +0.17 at $N = 12$), the FTP gains in one cell ($N = 8$: $0.926 \to 0.973$, $+17/-5$ discordant, $p = 0.017$), CE in two, MBLS in one, and GLS in none (all $p \geq 0.13$). The method *ordering* is unaffected: the FTP$-$GLS recovery margin is equal or wider on the denser grid ($0.20 \to 0.25$ at $N = 4$; 0.539 on both grids at $N = 8$), so the FTP-versus-GLS contrasts quoted here are conservative. A dense-grid closure rerun will finalize the absolute sparse-cell rates; every absolute rate in this section is quoted "at the 10^4-frequency production grid."

TABLE 2
Robustness arms (single seed, 256 sources per cell, $K = 2$, 10^4-frequency grid): period-recovery rate at $N = 4/8/12$ epochs per band. Holdout: truth shapes drawn from a held-out half of the library, so the vocabulary never contains the generating shapes. Cross-universe: truth from the RRab+RRc DECam universe, library and vocabularies from the Sesar universe. Band-amplitude: truth amplitudes differ by a factor 1.4 between g and r, violating the shared-amplitude model.

Arm	Method	N=4	N=8	N=12
holdout (50%)	FTP (PAM)	0.164	0.891	0.965
	FTP (greedy)	0.152	0.879	0.973
	GLS	0.047	0.449	0.898
cross-universe	FTP (PAM)	0.102	0.695	0.973
	FTP (greedy)	0.109	0.887	0.988
	GLS	0.086	0.672	0.926
band-amp. 1.4	FTP (PAM)	0.082	0.898	0.988
	FTP (greedy)	0.031	0.793	0.988
	GLS	0.047	0.328	0.867

5.3. *Recovery versus vocabulary size*

Figure 9 asks how large the vocabulary must be, in the regime where it matters (six epochs per band). On the RRab-dominated Sesar et al. (2010) universe the pooled PAM curve at $K = 1, 2, 4, 8$ is 0.716, 0.715, 0.738, 0.741 with mutually overlapping Wilson intervals; the paired K=1-versus-K=2 contrast is 34 vs. 33 discordant sources ($p = 1.0$) – the curve is flat in K within the sampling noise, and the smallest K whose interval overlaps the best cell's is $K = 1$. (A naive 98%-of-maximum rule would return $K = 4$; the differences it chases are not significant.) On the RRab+RRc universe of Baeza-Villagra et al. (2025) the same curve is 0.512, 0.703, 0.727, 0.719: the K=1 → 2 jump of +0.191 is highly significant (13 vs. 62 discordant, $p \approx 8 \times 10^{-9}$) and the curve is flat thereafter. The two panels together give the honest reading: *vocabulary size matters when the target population is morphologically diverse* – one medoid cannot serve both RRab and RRc – *and is a second-order effect for a shape-homogeneous class*, where the gain from $K = 1$ to $K = 8$ is at most ~ 0.03 and statistically marginal. Even at $K = 1$ every FTP point sits far above the sparse-regime baselines (GLS 0.171, MHLS 0.197, MBLS 0.090 on the Sesar universe; GLS 0.422 on the diverse universe). This recovery-rate flatness is complementary to the significance cost of Section 3.2, where enlarging K raises the catalog-max detection threshold: together they argue for the smallest K that covers the class morphology.

5.4. *Robustness arms and empirical photometric errors*

Table 2 stresses the three assumptions the headline setup makes. With half the library held out of vocabulary construction (truth shapes never available as templates), the sparse-regime advantage survives essentially intact (0.891 vs. 0.449 at $N = 8$): the vocabulary generalizes across shapes rather than memorizing them. With a 1.4× band-amplitude ratio – misspecifying the shared-amplitude signal model – the advantage again persists (0.898 vs. 0.328 at $N = 8$). The cross-universe arm is the hardest test and the most informative: with RRab+RRc truth but a Sesar-only library, the *unsupervised* PAM vocabulary degrades toward GLS (0.695 vs. 0.672 at $N = 8$, and one source *behind* GLS at $N = 16$, 0.996 vs. 1.000), while the *recovery-driven* greedy vocabulary – which sees a simulated population from the target universe during selection – adapts and retains the full advantage (0.887 at $N = 8$). When the library may not span the target population, the supervised selector is the safer choice.

A fourth arm replaces the synthetic noise model with the empirical ZTF error-versus-magnitude relation measured from the real light curves used for the cadence models of this paper's follow-up work; the empirical curve is a factor 1.4–3.5 larger than the synthetic one over its measured range ($14.3 \le m \le 18.5$). Recovery is essentially unchanged – FTP(PAM) 0.215/0.918/0.988 at $N = 4/8/12$ versus 0.233/0.902/0.984 pooled synthetic (this arm ran at $K = 4$; the Sesar curve is flat in K) – because RR Lyrae-scale amplitudes keep the per-epoch shape signal-to-noise high: in this regime recovery is limited by epoch count and the window function, not by photometric noise.

5.5. *Cost versus accuracy*

Finally, the same harness times the FTP against the non-linear template-fitting procedure it replaces: a Sesar-style oracle (cf. Sesar et al. 2010, 2016) that, at every (template, frequency) pair, brute-force scans 128 phase shifts, solves amplitude and per-band offsets linearly at each, and polishes the best phase by golden-section refinement. On a 32-source subsample at 1500 frequencies with the $K = 2$ vocabulary the two return *identical* recovered periods (recovery 0.344 for both) – consistent with the $< 10^{-6}$ agreement of their per-frequency optima – at 23.5 s for the FTP versus 56.2 s for the oracle, a measured ≈2.4× wall-time advantage under this harness's deliberately modest settings (128 phase trials, $N_{\rm obs} \le 120$). The ratio is structural rather than incidental: the oracle's per-frequency cost scales with the number of phase trials *and* with $N_{\rm obs}$, while the FTP's per-frequency cost is independent of $N_{\rm obs}$ (Section 2.3), so the advantage grows without bound with denser phase grids and longer light curves – the single-band measurements of Figure 6 reach $\gtrsim$480× by $N_{\rm obs} = 250$ against a generic non-linear optimizer. Reported end-to-end per-star costs of survey template-fitting pipelines are larger by further orders of magnitude, but those figures fold in denser phase grids, per-band template sets, and infrastructure external to the fitting algorithm, so we quote only the like-for-like in-harness ratio as a measured result.

6. DISCUSSION

Template fitting is a powerful technique for accurately recovering the period and amplitude of objects with *a priori* known lightcurve shapes. It has been used in the literature by, e.g. Stringer et al. (2019); Sesar et al. (2016, 2010), to detect RR Lyrae in the SDSS, PS1, and DES datasets, where it has been shown to produce purer samples of RR Lyrae at a given completeness. The computational cost of current template fitting algorithms, however, prevents their application to larger datasets.

We have presented a novel template fitting algorithm that extends the Lomb-Scargle periodogram (Lomb 1976; Scargle 1982; Barning 1963; Vaníček 1971) to handle non-sinusoidal signals that can be expressed in terms of a fixed truncated Fourier series with a reasonably small number of harmonics ($H \lesssim 10$).

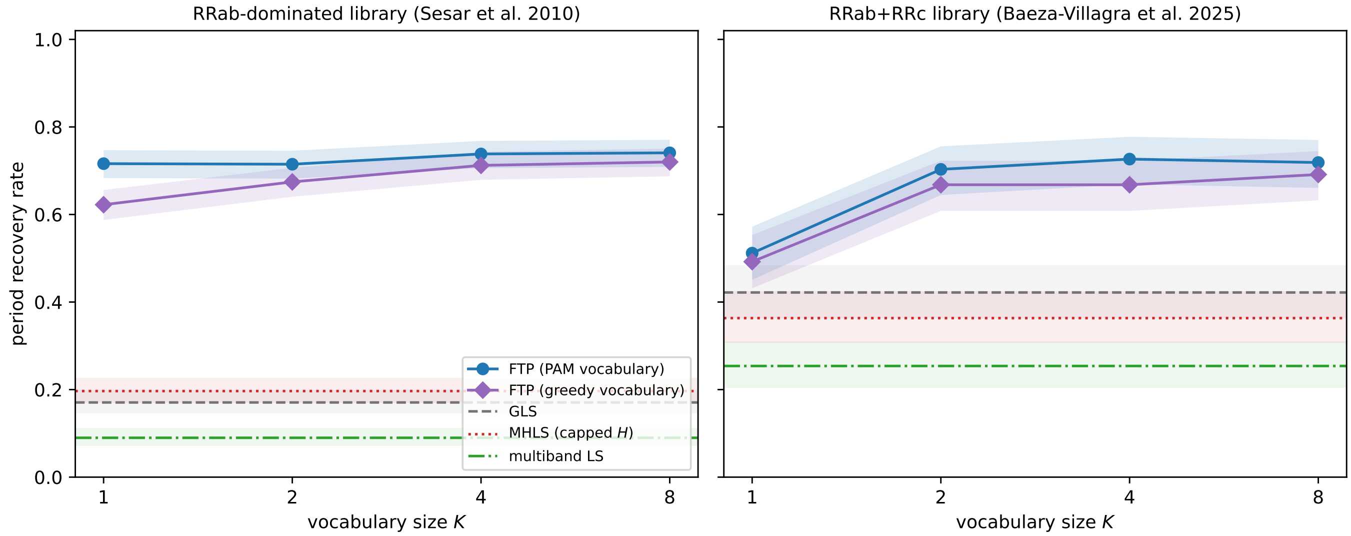

FIG. 9.— Sparse-regime (6 epochs per band) period recovery versus vocabulary size K for PAM and greedy vocabularies, with the K-independent baselines as reference bands. *Left*: the RRab-dominated Sesar et al. (2010) universe (768 pooled sources) is flat in K – a single well-chosen medoid already covers the population. *Right*: the morphologically diverse RRab+RRc universe of Baeza-Villagra et al. (2025) (256 sources) needs $K = 2$: the K=1 $\rightarrow$ 2 jump is +0.19 (McNemar $p \approx 8 \times 10^{-9}$), after which the curve is flat.

On its default scan path the fast template periodogram (FTP) asymptotically scales as $\mathcal{O}(N_f H^2 \log H N_f + N_f H^3)$ (including the extra factor of H needed to resolve a peak), while previous template fitting algorithms such as the one used in the `gatspy` library (VanderPlas 2016) scale as $\mathcal{O}(N_f N_{\rm obs})$. The FTP's cost is set by computing the polynomial coefficients and the per-frequency phase maximisation, and is independent of $N_{\rm obs}$; the reference root-finding path instead carries an $N_f H^4$ term. The remaining H^3 scaling restricts templates to those that are sufficiently smooth to be explained by a small number of Fourier terms.

FTP also improves the accuracy of previous template fitting algorithms, which rely on non-linear optimization at each trial frequency to minimize the V of the template fit. The FTP routinely finds superior fits over non-linear optimization methods.

An open-source Python implementation of the FTP is available at GitHub.[9] The current implementation could likely be improved by:

1. Further optimising the polynomial coefficient assembly and the scan+polish maximisation—for example by moving the per-chunk assembly onto vectorised or compiled kernels—beyond the measured factor of $\sim$25–35 that the scan+polish default already gains over the reference root-finding path.

2. Exploiting the embarassingly parallel nature of the FTP using GPU's.

3. Extending the vocabulary search of Sections 4–5 – which fits each template separately and reports the per-frequency maximum – to a *joint* shared-frequency model with independent amplitudes and phases per template. The polynomial-root formulation generalises to that setting analogously to the multi-band extension of Appendix B.

[9] https://github.com/PrincetonUniversity/FastTemplatePeriodogram

For a constant baseline, the default implementation improves on full non-linear optimization by more than two orders of magnitude ($\gtrsim 480\times$ measured by $N_{\rm obs} = 250$) once the sampling is well enough conditioned that the scan avoids its exact-root fallback, with the advantage growing as $N_{\rm obs}$ increases. This work presents the mathematical shortcut that, realised through the scan+polish maximiser, makes template fitting a fast and valuable tool for detecting non-sinusoidal periodic signals of specified shape in large time-series datasets.

A real-survey demonstration of the multi-band FTP against single-band Lomb–Scargle, multi-band Lomb–Scargle (VanderPlas & Ivezić 2015), and full non-linear template fitting (Sesar et al. 2016) is the natural next step and is the subject of follow-up work using the Zwicky Transient Facility public DR22 sparse multi-band photometry of RR Lyrae, with the Gaia DR3 RR Lyrae catalogue (Clementini et al. 2023) serving as an all-sky cross-matching benchmark.

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DATA AVAILABILITY

The `ftperiodogram` code implementing the fast template periodogram is openly available at https://github.com/PrincetonUniversity/FastTemplatePeriodogram. The manuscript sources, the figure-generation scripts, and the saved results underlying every figure and table in this article are available at https://github.com/PrincetonUniversity/FastTemplatePeriodogramPaper. The simulation-experiment artifacts – per-run `results.json` files with their full configuration blocks and random seeds – are versioned under `experiments/` in the code repository, so every quoted number regenerates from a committed script and recorded seed. The real Zwicky Transient Facility light curves used for the empirical cadence and photometric-error models were retrieved from the public IRSA ZTF light-curve service (`nph_light_curves`, DR22-era release); the cached light curves, including per-record query provenance and the observed MJD span, are stored alongside the experiment code.

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APPENDIX

FULL DERIVATION OF THE FAST TEMPLATE PERIODOGRAM

This appendix gives the explicit algebra summarised in Section 2. We present (i) the general form of the gradient conditions for a weighted-least-squares periodogram; (ii) the polynomial expansion of the phase-shifted template; (iii) the closed-form solution to the linear least-squares sub-problem at fixed phase shift θ_2; (iv) the reduction of the residual sum of squares V to the compact periodogram form of Equation 20; (v) the polynomial expansion of the weighted statistics MM and YM in terms of $\psi = e^{i\theta_2}$ and the trigonometric sums of Equations 23–27; and (vi) the algebraic identity converting the non-linear condition Equation 21 into the polynomial root-finding condition of Equation 22.

General form of the gradient conditions

The optimality condition Equation 17 can be written explicitly for each parameter θ_j as follows. Writing $\hat{y}_i = \hat{y}(\omega t_i;\theta)$ and $\partial_j \hat{y}_i = \partial \hat{y}(\omega t;\theta)/\partial\theta_j|_{t=t_i}$,

$$\begin{aligned} 0 &= \left.\frac{\partial V}{\partial \theta_j}\right|_{\theta=\theta_{\rm opt}} \\ &= -2\sum_i \mathsf{w}_i\,(y_i - \hat{y}_i)\,(\partial_j\hat{y})_i, \\ \sum_i \mathsf{w}_i y_i (\partial_j \hat{y})_i &= \sum_i \mathsf{w}_i \hat{y}_i (\partial_j\hat{y})_i. \end{aligned} \tag{A1}$$

This is a general result that extends to all least-squares periodograms. To simplify the derivations below we adopt the weighted-average notation

$$\langle X\rangle \equiv \sum_i \mathsf{w}_i X_i, \tag{A2}$$

$$\langle XY\rangle \equiv \sum_i \mathsf{w}_i X_i Y_i, \tag{A3}$$

$$\mathrm{Cov}(X,Y) \equiv \langle XY\rangle - \langle X\rangle\langle Y\rangle, \tag{A4}$$

$$\mathrm{Var}(X) \equiv \mathrm{Cov}(X,X). \tag{A5}$$

Phase-shifted template expansion

The transposed template $\mathbf{M}_{\theta_2} = \mathbf{M}(\omega t - \theta_2)$ can be expressed as

$$\begin{aligned} \mathbf{M}_{\theta_2}(\omega t) &= \sum_n c_n \cos n\,(\omega t - \theta_2) + s_n \sin n\,(\omega t - \theta_2) \\ &= \sum_n (c_n \cos n\theta_2 - s_n \sin n\theta_2)\cos n\omega t \\ &\qquad + (s_n \cos n\theta_2 + c_n \sin n\theta_2)\sin n\omega t \\ &= \sum_n \left(\alpha_n \psi^n + \alpha_n^* \psi^{-n}\right)\cos n\omega t \\ &\qquad - i\left(\alpha_n \psi^n - \alpha_n^*\psi^{-n}\right)\sin n\omega t \\ &= \sum_n A_n(\psi)\cos n\omega t + B_n(\psi)\sin n\omega t, \end{aligned} \tag{A6}$$

where $\alpha_n = (c_n + is_n)/2$ and $\psi \equiv e^{i\theta_2}$. We also define the weighted statistics

$$\begin{aligned} \widehat{YM} &\equiv \langle y\mathbf{M}_{\theta_2}\rangle, & \widehat{MM} &\equiv \left\langle \mathbf{M}_{\theta_2}^2\right\rangle, \\ \overline{M} &\equiv \langle \mathbf{M}_{\theta_2}\rangle, & MM &\equiv \mathrm{Var}(\mathbf{M}_{\theta_2}) = \widehat{MM} - \overline{M}^2, \\ YM &\equiv \mathrm{Cov}(\mathbf{M}_{\theta_2}, y) = \widehat{YM} - \bar{y}\overline{M}. \end{aligned} \tag{A7}$$

Closed-form linear least squares for (θ_1, θ_3)

For a given phase shift θ_2, the optimal amplitude and offset are obtained by setting the partial derivatives of V with respect to θ_1 and θ_3 to zero:

$$\begin{aligned} 0 = \frac{\partial V}{\partial\theta_1} &= -2\sum_i \mathsf{w}_i (y_i - \theta_1 \mathbf{M}_{\theta_2} - \theta_3)\mathbf{M}_{\theta_2} \\ &\propto \widehat{YM} - \theta_1\widehat{MM} - \theta_3\overline{M}, \end{aligned} \tag{A8}$$

$$\begin{aligned} 0 = \frac{\partial V}{\partial\theta_3} &= -2\sum_i \mathsf{w}_i (y_i - \theta_1 \mathbf{M}_{\theta_2} - \theta_3) \\ &\propto \bar{y} - \theta_1\overline{M} - \theta_3. \end{aligned} \tag{A9}$$

This 2×2 system can be written as

$$\begin{pmatrix} \widehat{MM} & \overline{M} \\ \overline{M} & 1 \end{pmatrix}\begin{pmatrix}\theta_1 \\ \theta_3\end{pmatrix} = \begin{pmatrix} \widehat{YM} \\ \bar{y}\end{pmatrix}, \tag{A10}$$

with closed-form solution

$$\begin{pmatrix}\theta_1\\ \theta_3\end{pmatrix} = \begin{pmatrix} YM/MM \\ \bar{y} - \overline{M}\,(YM/MM)\end{pmatrix}, \tag{A11}$$

where $MM = \widehat{MM} - \overline{M}^2$ and $YM = \widehat{YM} - \bar{y}\overline{M}$. The fitted model is then

$$\hat{y}_i = \bar{y} + \left(\frac{YM}{MM}\right)\left(M_i - \overline{M}\right). \tag{A12}$$

The periodogram in terms of YY, YM, and MM

Substituting Equation A11 into V,

$$\begin{aligned} V &= \sum_i \mathsf{w}_i (y_i - \hat{y}_i)^2 = \sum_i \mathsf{w}_i (y_i^2 - 2y_i\hat{y}_i + \hat{y}_i^2) \\ &= YY - 2\frac{(YM)^2}{MM} + \frac{(YM)^2}{MM} \\ &= YY - \frac{(YM)^2}{MM}. \end{aligned} \tag{A13}$$

Since $V_0 = YY$,

$$P(\omega) = 1 - \frac{V}{V_0} = \frac{(YM)^2}{YY \cdot MM}, \tag{A14}$$

which is Equation 20 of the body.

Polynomial expansion of MM and YM

The autocovariance of the template values, MM, is given by

$$\begin{aligned} MM &\equiv \sum_i \mathsf{w}_i M_i^2 - \left(\sum_i \mathsf{w}_i M_i\right)^2 \\ &= \sum_i \mathsf{w}_i \left(\sum_n A_n \cos\omega n t_i + B_n \sin\omega n t_i\right)^2 \\ &\quad - \left(\sum_n A_n C_n + B_n S_n\right)^2 \\ &= \sum_{n,m} A_n A_m CC_{nm} + 2A_n B_m CS_{nm} \\ &\quad + B_n B_m SS_{nm}, \end{aligned} \tag{A15}$$

using the symmetry $\sum_{n,m} A_n B_m CS_{nm} = \sum_{n,m} A_m B_n CS_{mn}$.

The products $A_n A_m$, $A_n B_m$, $B_n B_m$ expand as

$$\begin{aligned} A_n A_m &= \alpha_n\alpha_m\psi^{n+m} + \alpha_n^*\alpha_m\psi^{m-n} \\ &\quad + \alpha_n\alpha_m^*\psi^{n-m} + \alpha_n^*\alpha_m^*\psi^{-n-m}, \end{aligned} \tag{A16}$$

$$\begin{aligned} A_n B_m &= -i\big[\alpha_n\alpha_m\psi^{n+m} + \alpha_n^*\alpha_m\psi^{m-n} \\ &\quad - \alpha_n\alpha_m^*\psi^{n-m} - \alpha_n^*\alpha_m^*\psi^{-n-m}\big], \end{aligned} \tag{A17}$$

$$\begin{aligned} B_n B_m &= -\big[\alpha_n\alpha_m\psi^{n+m} - \alpha_n^*\alpha_m\psi^{m-n} \\ &\quad - \alpha_n\alpha_m^*\psi^{n-m} + \alpha_n^*\alpha_m^*\psi^{-n-m}\big]. \end{aligned} \tag{A18}$$

Substituting these into the MM sum gives the polynomial form

$$\begin{aligned} MM_{nm} &= A_n A_m CC_{nm} + 2A_n B_m CS_{nm} + B_n B_m SS_{nm} \\ &= \alpha_n\alpha_m\widetilde{CC}_{nm}\psi^{n+m} + 2\alpha_n\alpha_m^*\widetilde{CS}_{nm}\psi^{n-m} \\ &\quad + \alpha_n^*\alpha_m^*\widetilde{SS}_{nm}\psi^{-(n+m)}, \end{aligned} \tag{A19}$$

where

$$\widetilde{CC}_{nm} = (CC_{nm} - SS_{nm}) - i\left(CS + CS^T\right)_{nm}, \tag{A20}$$
$$\widetilde{CS}_{nm} = (CC_{nm} + SS_{nm}) + i\left(CS - CS^T\right)_{nm}, \tag{A21}$$
$$\widetilde{SS}_{nm} = (CC_{nm} - SS_{nm}) + i\left(CS + CS^T\right)_{nm}. \tag{A22}$$

Analogously, the cross term YM expands as

$$\begin{aligned} YM_k &= A_k YC_k + B_k YS_k \\ &= \alpha_k YC_k \psi^k + \alpha_k^* YC_k \psi^{-k} \\ &\quad - i\left(\alpha_k YS_k \psi^k - \alpha_k^* YS_k \psi^{-k}\right) \\ &= \alpha_k \widetilde{YC}_k \psi^k + \alpha_k^* \widetilde{YC}_k^* \psi^{-k}, \end{aligned} \tag{A23}$$

where $\widetilde{YC}_k \equiv YC_k - iYS_k$.

From the non-linear condition to the polynomial form

The non-linear condition Equation 21 can be rewritten in terms of the rescaled quantities $YM' \equiv \psi^H YM$ and $MM' \equiv \psi^{2H} MM$, which (from Equations A19–A23) are polynomials in ψ of degree $2H$ and $4H$, respectively. Their derivatives are

$$\partial YM' = H\psi^{H-1} YM + \psi^H \partial YM, \tag{A24}$$
$$\partial MM' = 2H\psi^{2H-1} MM + \psi^{2H} \partial MM. \tag{A25}$$

Combining these,

$$\begin{aligned} 0 &= 2MM\,\partial(YM) - YM\,\partial(MM) \\ &= \psi^{3H+1}\left[2MM\,\partial(YM) - YM\,\partial(MM)\right] \\ &= 2\,\psi^{2H} MM\left[\psi^{H+1}\partial(YM)\right] \\ &\quad - \psi^H YM\left[\psi^{2H+1}\partial(MM)\right] \\ &= 2\,MM'\left[\psi\,\partial(YM') - H\,(YM')\right] \\ &\quad - YM'\left[\psi\,\partial(MM') - 2H\,(MM')\right] \\ &= \psi\left[2\,MM'\,\partial(YM') - YM'\,\partial(MM')\right] \\ &\quad - 2H\left[MM'\,YM' - YM'\,MM'\right] \\ &= 2\,MM'\,\partial(YM') - YM'\,\partial(MM'). \end{aligned} \tag{A26}$$

The last step assumes $\psi \neq 0$, which holds since $\psi = e^{i\theta_2}$ lies on the unit circle for all real θ_2. Equation A26 is the polynomial root-finding condition of Equation 22 in the body. Although the nominal degree of Equation A26 is $6H - 1$ ($\deg MM' = 4H$, $\deg \partial(YM') = 2H - 1$), the degree-$(6H{-}1)$ terms cancel identically: writing ym and mm for the leading coefficients of YM' and MM', the leading coefficient of $2\,MM'\,\partial(YM')$ is $2\,(2H)\,mm\,ym$ and that of $YM'\,\partial(MM')$ is $(4H)\,ym\,mm$ (a consequence of $\deg MM' = 2\deg YM'$), so the true degree is at most $6H - 2$.

MULTI-BAND TEMPLATE PERIODOGRAMS AND THE SHARING HIERARCHY

The single-band template periodogram extends to observations in K filters (N_k observations in the k-th filter, the i-th of which is denoted $y_i^{(k)}$) as a family of models that differ in which of the template's amplitude, phase, and offset are held common across the bands. We fit a periodic, multi-band template $\mathbf{M} = (\mathbf{M}^{(1)}, \mathbf{M}^{(2)}, ..., \mathbf{M}^{(K)}) : [0, 2\pi) \to \mathbb{R}^K$ to all observations, and label the members of the resulting *sharing hierarchy* by their names in the accompanying implementation (Table 3). Freeing more parameters per band increases the model's flexibility at the price of a larger per-frequency solve. The three modes that retain a per-band offset (`independent`, `shared_phase`, `floating_offsets`) measure signal power against the per-band-centred variance $\sum_k W^{(k)} YY^{(k)}$, whereas the fully shared `sesar` model centres on the global mean; every member reduces to the single-band periodogram of Appendix A at $K = 1$. The multi-band experiments in this paper use `floating_offsets` (shared amplitude and phase, per-band free offset), the default mode, which we derive in Section B.2 after the fully shared model.

The fully shared model (`sesar`)

In the model of Sesar et al. (2016) the amplitude, phase, and offset are all shared across the bands, up to a fixed, known per-band offset $\lambda^{(k)}$:

TABLE 3

Members of the multi-band sharing hierarchy, labelled by their name in the accompanying implementation. Each mode fixes some of the template's amplitude (θ_1), phase (θ_2), and offset (θ_3) across the K bands and frees the rest per band. V_0 is the reference variance that normalises the power (Equation 20); $W^{(k)} = \sum_i \mathsf{w}_i^{(k)}$ is the fraction of the total inverse-variance weight in band k ($\sum_k W^{(k)} = 1$), and $\bar{y}$ is the global weighted mean (computed on the $\lambda^{(k)}$-subtracted data for `sesar`). The final column is the per-frequency cost of the phase maximisation, on top of the $\mathcal{O}(KN_f H \log N_f H)$ NFFT summations that are common to every mode (H is the harmonic order). All modes coincide with the single-band periodogram at $K = 1$.

Mode (`code`)	Shared across bands	Free per band	Reference variance V_0	Root cost / frequency
`independent`	(nothing)	$\theta_1, \theta_2, \theta_3$	$\sum_k W^{(k)}\, YY^{(k)}$	$\mathcal{O}(KH^3)$
`shared_phase`	θ_2	θ_1, θ_3	$\sum_k W^{(k)}\, YY^{(k)}$	$\mathcal{O}((HK)^3)$
`floating_offsets`†	θ_1, θ_2	θ_3	$\sum_k W^{(k)}\, YY^{(k)}$	$\mathcal{O}(H^3)$
`sesar`	$\theta_1, \theta_2, \theta_3$ (fixed $\lambda^{(k)}$)	—	$\sum_{k,i} \mathsf{w}_i^{(k)} (y_i^{(k)} - \lambda^{(k)} - \bar{y})^2$	$\mathcal{O}(H^3)$

† Default mode used for the multi-band experiments in this paper. `independent` is the per-band template analogue of the VanderPlas & Ivezić (2015) multi-band periodogram; `sesar` is the model of Sesar et al. (2016).

$$\hat{y}^{(k)}(t;\theta) = \theta_1 \mathbf{M}^{(k)}(\omega t - \theta_2) + \theta_3 + \lambda^{(k)} \tag{B1}$$

where $\lambda^{(k)}$ is a fixed relative offset for band k. The V for this model is

$$V = \sum_{k=1}^{K} \sum_{i=1}^{N_k} \mathsf{w}_i^{(k)} \left(y_i^{(k)} - \hat{y}_i^{(k)} \right)^2 \tag{B2}$$

To make things simpler, we can set the $\lambda^{(k)}$ values to 0 simply by subtracting them off from our observations; this means we take all $y_i^{(k)} \to y_i^{(k)} - \lambda^{(k)}$. We have that

$$V = \sum_{k=1}^{K} W^{(k)} V_k \tag{B3}$$

Where $W^{(k)} \equiv \sum_{i=1}^{N_k} \mathsf{w}_i^{(k)}$ and $\sum_{k=1}^{K} W^{(k)} = 1$. This means that the system of equations reduces to:

$$0 = \frac{\partial V}{\partial \theta_1} = \sum_{k=1}^{K} W^{(k)} \left(\widehat{YM}^{(k)} - \theta_1 \widehat{MM}^{(k)} - \theta_3 \overline{M}^{(k)} \right) \tag{B4}$$

for the θ_1 parameter, where $\widehat{YM}^{(k)}$, $\widehat{MM}^{(k)}$, $\overline{M}^{(k)}$ are values from the single band case computed for each band individually, holding $W^{(k)} = 1$ for each band. That is:

$$\widehat{YM}^{(k)} \equiv \frac{1}{W^{(k)}} \sum_{i=1}^{N_k} \mathsf{w}_i^{(k)} y_i^{(k)} \mathbf{M}_{\theta_2}^{(k)}(\omega t_i) \tag{B5}$$

$$\widehat{MM}^{(k)} \equiv \frac{1}{W^{(k)}} \sum_{i=1}^{N_k} \mathsf{w}_i^{(k)} \left(\mathbf{M}_{\theta_2}^{(k)}(\omega t_i) \right)^2 \tag{B6}$$

$$\overline{M}^{(k)} \equiv \frac{1}{W^{(k)}} \sum_{i=1}^{N_k} \mathsf{w}_i^{(k)} \mathbf{M}_{\theta_2}^{(k)}(\omega t_i) \tag{B7}$$

For the offset θ_3, we have

$$0 = \frac{\partial V}{\partial \theta_3} = \sum_{k=1}^{K} W^{(k)} \left(\bar{y}^{(k)} - \theta_1 \overline{M}^{(k)} - \theta_3 \right) \tag{B8}$$

Where $\bar{y}^{(k)}$ is the weighted-mean for the k-th band ($\bar{y}^{(k)} \equiv (1/W^{(k)}) \sum_{i=1}^{N_k} \mathsf{w}_i^{(k)} y_i^{(k)}$), again with $y_i^{(k)} \to y_i^{(k)} - \lambda^{(k)}$. So if we redefine the quantities $\widehat{YM}$, $\widehat{MM}$, $\overline{M}$, and $\bar{y}$ as weighted averages across the bands, i.e. $\widehat{YM} = \sum_{k=1}^{K} W^{(k)} \widehat{YM}^{(k)}$, etc., the solution for the optimal parameters has the same form:

$$\theta_1 = \left(\frac{YM}{MM}\right) \tag{B9}$$

$$\theta_3 = \bar{y} - \overline{M}\left(\frac{YM}{MM}\right) \tag{B10}$$

which means that the model for the k-th band is

$$\hat{y}^{(k)} = \bar{y} + \left(\frac{YM}{MM}\right)\left(M_i^{(k)} - \overline{M}\right) \tag{B11}$$

The form of the periodogram has the same form as the single-band case:

$$V = \sum_{k=1}^{K}\sum_{i=1}^{N_k} \mathsf{w}_i^{(k)}\left(y_i^{(k)} - \hat{y}_i^{(k)}\right)^2 \tag{B12}$$

$$= \sum_{k=1}^{K}\sum_{i=1}^{N_k} \mathsf{w}_i^{(k)}\left((y_i^{(k)} - \bar{y}) - \left(\frac{YM}{MM}\right)(M_i^{(k)} - \overline{M})\right)^2 \tag{B13}$$

$$= \sum_{k=1}^{K}\sum_{i=1}^{N_k} \mathsf{w}_i^{(k)}\left((y_i^{(k)} - \bar{y})^2 - 2\left(\frac{YM}{MM}\right)(M_i^{(k)} - \overline{M})(y_i^{(k)} - \bar{y}) + \left(\frac{YM}{MM}\right)^2(M_i^{(k)} - \overline{M})^2\right)^2 \tag{B14}$$

$$= YY - 2\frac{(YM)^2}{MM} + \frac{(YM)^2}{MM} \tag{B15}$$

$$= YY - \frac{(YM)^2}{MM}. \tag{B16}$$

Since there is a single shared offset between the bands (i.e. we assume the mean magnitude is the same in all bands after subtracting $\lambda^{(k)}$), the variance for the signal, YY, is not $\sum_{k=1}^{K}\sum_{i=1}^{N_k}\mathsf{w}_i^{(k)}(y_i^{(k)} - \bar{y}^{(k)})^2$ but $\sum_{k=1}^{K}\sum_{i=1}^{N_k}\mathsf{w}_i^{(k)}(y_i^{(k)} - \bar{y})^2$.

Construction of the polynomial for the multi-band case can be performed by first computing the polynomial expression for $\psi^{2H}\widehat{MM} = \sum_{k=1}^{K}\sum_{i=1}^{N_k}\mathsf{w}_i^{(k)}\left(\psi^H M_i^{(k)}\right)^2$ and a separate polynomial expression for $\psi^H\overline{M}$, which can then be squared and subtracted from $\psi^{2H}\widehat{MM}$ to find $MM' = \psi^{2H}MM$.

For $YM' = \psi^H YM$, we merely compute $\psi^H\widehat{YM}^{(k)}$ for each band, take the weighted average of the polynomial coefficients for all bands ($\psi^H\widehat{YM} = \sum_k^K W^{(k)}\psi^H\widehat{YM}^{(k)}$) and subtract the $\psi^H\bar{y}\overline{M}$ polynomial to get YM'.

After computing the polynomials MM' and YM', the polynomial in Equation 22 can be computed quickly and the following steps for finding the optimal model parameters is the same as in the single band case.

Floating per-band offsets (`floating_offsets`)

The mode used for all of the multi-band experiments in this paper keeps the shared amplitude θ_1 and phase θ_2 of the `sesar` model but lets each band carry its own free offset $\theta_3^{(k)}$:

$$\hat{y}^{(k)}(t;\theta) = \theta_1\mathbf{M}^{(k)}(\omega t - \theta_2) + \theta_3^{(k)}. \tag{B17}$$

Writing $V = \sum_k W^{(k)}V_k$ as before, the offset stationarity condition now acts on each band *separately*,

$$0 = \frac{1}{W^{(k)}}\frac{\partial V}{\partial\theta_3^{(k)}} = \bar{y}^{(k)} - \theta_1\overline{M}^{(k)} - \theta_3^{(k)}, \tag{B18}$$

so the optimal offset $\theta_3^{(k)} = \bar{y}^{(k)} - \theta_1\overline{M}^{(k)}$ centres band k on its *own* weighted mean $\bar{y}^{(k)}$. Substituting back, band k's contribution is exactly the single-band, self-centred residual, and V separates into a weighted average of single-band forms,

$$V = \sum_{k=1}^{K} W^{(k)}\left(YY^{(k)} - 2\theta_1\,YM^{(k)} + \theta_1^2\,MM^{(k)}\right) = \overline{YY} - 2\theta_1\,\overline{YM} + \theta_1^2\,\overline{MM}, \tag{B19}$$

where $YM^{(k)} \equiv \widehat{YM}^{(k)} - \bar{y}^{(k)}\,\overline{M}^{(k)}$ and $MM^{(k)} \equiv \widehat{MM}^{(k)} - \left(\overline{M}^{(k)}\right)^2$ are the *centred* single-band moments YM and MM of Appendix A evaluated within band k—that is, computed about that band's own weighted mean, the same per-band objects that appear in Section B.3—$YY^{(k)}$ is the corresponding per-band-centred variance, and the band-combined quantities, written with an overline, are the *plain* $W^{(k)}$-weighted band averages

$$\overline{YM} = \sum_{k=1}^{K} W^{(k)}\, YM^{(k)}, \qquad \overline{MM} = \sum_{k=1}^{K} W^{(k)}\, MM^{(k)}, \qquad \overline{YY} = \sum_{k=1}^{K} W^{(k)}\, YY^{(k)}. \tag{B20}$$

(The hat remains reserved for the *uncentred* per-band moments $\widehat{YM}^{(k)}$, $\widehat{MM}^{(k)}$ of Section B.1, and the prime for the ψ-rescaled polynomials of Appendix A, e.g. $\overline{YM}' = \psi^H\, \overline{YM}$.) Because every band is already centred on its own mean, no global re-centring is required—unlike the `sesar` model (Section B.1), whose single shared offset forces the between-band $\bar{y}\overline{M}$ and $\overline{M}^2$ corrections. Minimising over the shared amplitude gives $\theta_1 = \overline{YM}/\overline{MM}$ and

$$P(\omega) = 1 - \frac{V}{\overline{YY}} = \frac{\left(\overline{YM}\right)^2}{\overline{YY}\;\overline{MM}}, \tag{B21}$$

the single-band closed form of Equation 20 with the band-averaged quantities in place of YM, MM, and YY. Since band averaging is linear in the per-band coefficient vectors, the rescaled polynomials $\overline{YM}' = \psi^H\, \overline{YM}$ and $\overline{MM}' = \psi^{2H}\, \overline{MM}$ have the same degrees ($2H$ and $4H$) as their single-band counterparts, so the maximisation over θ_2 uses the identical degree-$(6H-2)$ root condition of Equation A26. Floating per-band offsets therefore cost essentially one single-band solve per frequency, independent of K beyond the summations; at $K = 1$ ($W^{(k)} = 1$, $\bar{y}^{(k)} = \bar{y}$) the mode collapses to the single-band periodogram.

Shared phase, per-band amplitude (`shared_phase`)

Relaxing the shared amplitude of `floating_offsets` yields the `shared_phase` model: a single phase θ_2 is common to all bands, but each band fits its own amplitude and offset. For a fixed shared phase every band fits independently, so the variance explained is a sum of the per-band single-band contributions,

$$F(\psi) = \sum_{k=1}^{K} W^{(k)}\, \frac{\left(YM^{(k)}(\psi)\right)^2}{MM^{(k)}(\psi)}, \tag{B22}$$

and the power is $P(\omega) = \max_{\theta_2} F(\psi)/\overline{YY}$, normalised by the same per-band-centred variance $\overline{YY} = \sum_k W^{(k)} YY^{(k)}$ (Equation B20) as `floating_offsets`. The per-band amplitudes no longer collapse into a single band-combined polynomial. Clearing the common denominator $\prod_l \left(MM^{(l)}\right)^2$ (positive on the unit circle) gives the stationarity polynomial

$$G(\psi) = \sum_{k=1}^{K} W^{(k)}\, YM^{(k)}\, q_k \prod_{l \neq k} \left(MM^{(l)}\right)^2 = 0, \qquad q_k = 2\, MM^{(k)} \partial YM^{(k)} - \partial MM^{(k)}\, YM^{(k)}, \tag{B23}$$

whose degree is at most $8HK - 2$: each term's nominal leading coefficient at degree $8HK - 1$ cancels identically, by the same top-term identity that reduces the single-band q_k. Its roots must be found explicitly, at a cost of $\mathcal{O}((HK)^3)$ per frequency. `shared_phase` is thus the only member of the hierarchy whose phase maximisation *couples* the bands into a single polynomial of K-dependent degree, making its per-frequency cost superlinear in K; the `independent` mode's $\mathcal{O}(KH^3)$ cost (Table 3) is instead just K repetitions of the single-band solve. Because the shared phase couples the bands, the per-band amplitudes cannot each be independently forced non-negative; this mode returns the power-maximising shared phase and its fitted per-band amplitudes.

Computational cost across the hierarchy

For every mode the per-band NFFT summations cost $\mathcal{O}(KN_f H \log N_f H)$ over the full frequency grid, independent of the sharing mode. The per-frequency phase maximisation is where the modes differ. The coupled-but-cheap members `floating_offsets` and `sesar` each assemble a single band-combined polynomial of degree at most $6H - 2$ (Equation B20 and the `sesar` construction of Section B.1) and solve it at the single-band cost $\mathcal{O}(H^3)$ per frequency, so their total cost $\mathcal{O}(KN_f H \log N_f H + N_f H^3)$ exceeds a single-band periodogram's only through the K-fold NFFT summations; `independent` runs K such solves, at $\mathcal{O}(KN_f H^3)$. Only `shared_phase` is genuinely more expensive: its degree-$(8HK-2)$ stationarity polynomial G costs $\mathcal{O}((HK)^3)$ per frequency. The implementation accompanying this paper replaces that root-finding with a direct scan of $F(\psi)$ on $\mathcal{O}(HK)$ trial phases, forming and solving G only at those frequencies where *some* band's template-variance polynomial $|MM^{(k)}|$ nearly vanishes on the unit circle (its minimum over the circle falling below a fixed fraction of its maximum). This removes the $\mathcal{O}((HK)^3)$ cost away from such deep-$|MM|$ dips, but not at them, where the exact root path is still run and the better of the two solutions kept, so the saving is opportunistic rather than guaranteed. The realised speedup is therefore strongly configuration-dependent: on well-conditioned dense-grid `shared_phase` data ($K = 2$) we measure the scan to be 1.3× faster than the reference root path at $H = 3$ and 5.4× at $H = 8$, but on sparse or otherwise deep-$|MM|$ sampling, where the fallback

TABLE 4
ALIAS-RESOLVED RE-SCORING OF THE POOLED SESAR ET AL. (2010)-UNIVERSE RECOVERY RESULTS (768 SOURCES PER CELL, $K = 2$) UNDER PHASE COHERENCE $|\Delta f|\,T < 0.5$. COUNTS PER CELL: EXACT RECOVERIES, HARMONIC ALIASES, $\pm 1\,\mathrm{YR}^{-1}$ WINDOW BEATS, AND MISSES. ($P/3$ AND $\pm 1\,\mathrm{DAY}^{-1}$ ENTRIES ARE ZERO IN EVERY CELL SHOWN AND ARE OMITTED.)

N	Method	exact	$P/2$	$2P$	$3P$	yr beat	miss
8	FTP (PAM)	689	0	0	0	0	79
8	FTP (greedy)	666	0	0	0	1	101
8	GLS	307	0	1	0	1	459
8	MHLS	428	0	15	6	1	318
8	MBLS	184	1	0	0	0	583
8	CE	60	1	13	3	0	691
16	FTP (PAM)	768	0	0	0	0	0
16	FTP (greedy)	766	0	0	0	0	2
16	GLS	763	0	0	0	0	5
16	MHLS	595	0	70	23	1	79
16	MBLS	739	0	0	0	0	29
16	CE	643	1	39	1	0	84
40	FTP (PAM)	768	0	0	0	0	0
40	FTP (greedy)	768	0	0	0	0	0
40	GLS	768	0	0	0	0	0
40	MHLS	667	0	83	16	0	2
40	MBLS	768	0	0	0	0	0
40	CE	762	0	6	0	0	0

fires at most frequencies, the cost approaches that of the reference and the speedup collapses toward $1\times$. Since the polynomial step dominates in most applications, the multi-band template periodogram in the shared-amplitude modes is not significantly more computationally intensive than the single-band case.

ALIAS BREAKDOWN OF THE SIMULATED RECOVERY RATES

The headline recovery criterion of Section 5 – $|P_{\rm rec} - P_{\rm true}|/P_{\rm true} < 0.01$, exact only – has one known blind spot on this search band: a $\pm 1\,\mathrm{yr}^{-1}$ window beat displaces the frequency by only $|\Delta f| = 1/365.25\,\mathrm{cyc\,day}^{-1}$, a fractional period error of at most 0.27% for $f \geq 1\,\mathrm{cyc\,day}^{-1}$, so the fractional criterion cannot distinguish a year beat from an exact recovery. Every scored (method, cell) pair therefore persists its per-source recovered periods, and Table 4 re-scores them under baseline-aware phase coherence, $|\Delta f|\,T < 0.5$ (the accumulated-phase-drift convention of Graham et al. 2013b): over the $T = 1095.75\,\mathrm{d}$ baseline a year beat accumulates $|\Delta f|\,T = 3$ cycles of drift and is cleanly separated, while each non-exact recovery is assigned to the nearest member of a named alias set (harmonic multiples $P/2$, $2P$, $3P$ and the $\pm 1\,\mathrm{day}^{-1}$, $\pm 1\,\mathrm{yr}^{-1}$ window beats).

Three facts emerge. First, the headline is safe: year-beat contamination is at most one source per cell (and $\pm 1\,\mathrm{day}^{-1}$ beats are entirely absent at $N \geq 8$), and the two scoring conventions agree on the headline rates – e.g. FTP (PAM) at $N = 8$ is $689/768 = 0.897$ phase-coherent-exact versus 0.902 under the fractional criterion. Second, the FTP's sparse-regime failures are genuine misses, not aliases: at $N = 8$ none of its 79 failures land on a named alias. Third, the MHLS plateau of Figure 8 is harmonic aliasing, not noise: at $N = 40$ its $2P$ and $3P$ aliases account for 99/768 sources, and crediting them would lift MHLS from 0.868 to 0.997 – the free-shape fit detects *a* periodicity but, lacking a shape prior, often locks onto a harmonic of the fundamental. The conditional-entropy statistic shows the same signature at lower amplitude. The full breakdown over all epoch counts regenerates from the committed per-source artifacts and scripts (see Data Availability).